Adagio Institute, Inc.
Houston, Texas, United States

# Doctoral Examination Report

## Technical Annex: Examination Record

Examination for the Degree of
Doctor of Philosophy in Mathematical Physics
(Concentration in Stochastic Quantum Theory)
by Published Works

| | |
|---|---|
| **Candidate:** | Benjamin D. Summers |
| **Doctoral Examiner:** | E. F. Boero<br>Director, Adagio Institute, Inc. |
| **Independent evaluation:** | Dr S. R. Ismail-Sutton, Durham University |
| **Report date:** | October 2, 2026 |

This Technical Annex accompanies the Doctoral Examination Report ("the Report") and forms part of the examination record. It sets out, paper by paper, the findings of the two technical reviews, the candidate's responses embedded beneath the corresponding findings, and the Doctoral Examiner's dispositions, together with the record of rounds, the corrections to the examination record, and the full record of revisions. The Report states the assessment and the recommendation; this Annex is the evidence on which they rest. This Annex, the Report and the exhibits are held in the Institute's GitHub repository, https://github.com/Adagio-Group/Adagio-Institute, which also keeps the record of the examination process; the Annex and the exhibits form part of the Doctoral Examination Report for the purposes of Section 5.3 of the regulations. References to sections of the Report are given as "Report, Section $n$".

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## T.1 Form and rounds of the examination

Every paper was examined in the same form: an itemised technical report by the Doctoral Examiner (numbered Findings, each classified, and numbered Questions), an independent technical evaluation by Dr S. R. Ismail-Sutton (numbered or itemised points), a written response by the candidate to both, and further rounds of correspondence in which the Doctoral Examiner adjudicated the responses against the revised text. The two reviews were prepared without conferring; on Papers 3–6 they converged on the same principal points, as the candidate noted in his responses (for Paper 5, although the two reviews had examined different texts). Table T.1 records the rounds. The review correspondence is deposited in full as Exhibit A (09_procedural_correspondence) rather than paraphrased, so that every exchange summarized in this report can be assessed directly. Each message is filed as COR-P*n*-*k*, numbered by paper and in date order, with the date, the sender and recipient (CAN, the candidate; EX01, the Doctoral Examiner; EX02, the independent evaluator; EXAMINERS, both) and the type of document in the file name; the register of each paper's correspondence closes its section. The Doctoral Examiner acted throughout as the technical examiner of each paper as well as the recommending examiner; this dual role is disclosed in the Report, Section 6.

**How to read the paper sections.** Each of Sections T.3–T.8 follows the same unit: the technical examiner's verdict and the candidate's global response; then each finding, with its severity, the candidate's response and the Doctoral Examiner's disposition; the corrections the candidate identified himself; the corrected architecture of the paper; the classification of results; and the register of the correspondence on the paper. Findings carry the numbers of the reports as transmitted, which are the numbers the candidate's letters cite. "The examiner" is the Doctoral Examiner acting as technical examiner; "the independent evaluator" is Dr S. R. Ismail-Sutton.

## T.2 Corrections to the examination record

The following errors and omissions of the examination itself were identified in the course of the review, several of them by the candidate. None alters the recommendation; each is corrected here, and the earlier documents should be read subject to this list.

(a) *Paper 5, text examined.* The Doctoral Examiner's first-round report on Paper 5 cited the repository DOI but examined, at every checkable point, the text of 14 July 2026 rather than the later July deposit then current (which the independent evaluator reviewed), owing to a July-dated file and a mistyped DOI. The candidate states that Findings 2–7, two local errors (the Moreau decomposition of Section 6 and the bound of Proposition 7.1) and several presentational items had already been addressed in that deposit. The Doctoral Examiner acknowledged the mismatch in writing (22 September) and re-established that his verification of the new material refers to revision 2.1. The material items of the portfolio-level evaluation on Paper 5, Section 8 (the non-monotone Table 1), and the supplementary prior-art report likewise describe the 14-July text. The numerical section has since been withdrawn entirely, so the outcome is unaffected.

(b) *Paper 5, Remark 2.3.* The first-round report endorsed the reading of equation (5) as $\delta_V \Delta_V = 0$ "for the displayed twisted coboundary". The display examined had the two twisted terms with the wrong sign, so that $\delta_V \Delta_V$ reduces to twice a difference of two deficits (verified for this

report against the 14-July text). The candidate pointed this out; the corrected display satisfies the identity, and the version of record adds the formula for $\delta_V^2$.

(c) *Paper 2, origin of three corrections.* The portfolio-level evaluation (Section 3) and the supplementary report credit the candidate with independently finding and correcting the unsatisfiability of hypothesis (C2) of Theorem 7.4 in infinite dimensions, the per-pair minimality in Theorem 5.4(iv), and the witness of Proposition 2.11(c) for $n > 3$. The candidate's letter of 3 September shows that all three were diagnosed in the independent evaluator's review (her items 9, 8 and 1); the repairs as written are the candidate's. The evidence of independent research ability in the Report, Section 3.2, is drawn from elsewhere in the record.

(d) *Paper 2, Appendix C.* The misapplication of Michael's selection theorem in Step 1b of the completion argument was not detected by the Doctoral Examiner's audit; the independent evaluator found it.

(e) *Paper 2, Finding 4.* The Doctoral Examiner's finding that the constant $1/3$ is optimal offered an "extremal family" with "equality requiring both intermediate kernels to be deterministic", which implies attainability. The candidate's Proposition 4.7 shows the constant is optimal only as an infimum, not attained. Corrected by the Doctoral Examiner in writing (25 September); his derivation was a first-order argument in the symmetric two-state reduction.

(f) *Paper 1, first-round report.* The report reached the candidate (11 September) with an unresolved cross-reference in Question 5 ("Finding ??"), without the verification section it cites twice and the counterexample recorded there, and dated 4 September while examining the version of 7 September; the date had been carried over from the other reports of the batch. The candidate identified these three defects (14 September). In the second round (26 September) the Doctoral Examiner supplied the missing material and withdrew two prescriptions that the revision of 14 September had adopted. Finding 5 had recommended the reading on which "the sharp surface itself carries no algebra", and Section 4 had concluded from it that the surface does not carry the determination; that does not follow, because the Haag–Kastler net is indexed by open regions and a sharp Cauchy surface has empty interior, so the absence of an algebra is a feature of the indexing. Question 1 had asked for "well-posed evolution", and Definition 1(a) said "well posed" without continuous dependence on the data. Both are corrected in the version of record. The corrected report, TR-P01-DOCv2-EX01-Rv2, was sent on 29 September; its object remains the version of 7 September, and it is the report deposited (Exhibit I).

(g) *Paper 1, the open problem.* In the second round the Doctoral Examiner withdrew a statement of his working record that the later papers supply a description-independent invariant (the deficit with its intrinsic locus and $\beta$, the liftability profile, the memory rank): those quantities are invariants relative to a transition family and an observed space already fixed. He posed the open problem instead as the absence of a description-independent invariant. The candidate showed that no such invariant can exist, since augmenting the state with its full history renders any process Markov, and restated the problem relative to a physically fixed class of admissible descriptions; the Doctoral Examiner accepted the correction (27 September). The problem stands in that form in Section 7 of Paper 1 and in the Report, Section 3.1.

(h) *Paper 3, re-entry point.* The Doctoral Examiner's second-round value 0.4885 for the re-entry coupling at $\varepsilon = 0.01$ was computed with an interaction scale taken from an unresolved grid of triples; the closed form gives 0.4882, which the manuscript prints. Corrected by the candidate (24 September) and acknowledged by the Doctoral Examiner (27 September).

(i) *Paper 4, Finding 2.* The Doctoral Examiner's statement of the product identity gave $1/\sigma^2_{\text{eff}} = 1/\sigma^2_{\Pi} + \lambda/\sigma^2_{\nu}$ as exact. It holds to leading order in the grid step; the exact interior-row relation is in $k^2/\sigma^2$. Corrected by the candidate and accepted by the Doctoral Examiner in writing (23 September).

(j) *Paper 4, Monte Carlo figures.* The Doctoral Examiner's first-round value 3.44 for $\sqrt{N} \cdot \text{std}$ at $N = 10^4$ (700 replications), and the single-batch value 7.53, could not be reproduced: the candidate obtained $4.02 \pm 0.05$ on 10,000 replications, the Doctoral Examiner's own third run gave $3.817 \pm 0.127$, and an independent run for this report (4,000 replications, fresh seed) gave $3.97 \pm 0.08$, consistent with the paper's Table 1 ($3.95 \pm 0.14$). Resampling 700 replications from the independent run gives a value at or below 3.44 with probability below 0.001 (the candidate's run gives under $3 \times 10^{-4}$). In his letter of 23 September the Doctoral Examiner accepted 7.53 as an artefact of his implementation and left 3.44 undecided, on the ground that the three estimates differed by more than their standard errors; the candidate showed (24 September) that this holds only for the gap between 3.44 and 4.02, the other two runs being compatible. Both figures are withdrawn from the verification record. The conclusions drawn from them (the statistic lies well below its asymptote at $N = 10^4$; most plug-in $\hat{\chi}^2$ draws fall below the exact value) hold at the reproduced figures.

(k) *Paper 4, the backward recursion.* In his letter of 23 September the Doctoral Examiner stated that the recursion in the first proof of Proposition 7.5, run from the last index backward, yields variances conditional on the future and not the innovation variances of the forward representation. The candidate showed that it yields the forward innovation variances (24 September), and the Doctoral Examiner withdrew the observation after verifying it on a chain with $M = 12$ (28 September). The proof was replaced for a different reason (Section T.6, Finding 4).

(l) *Paper 4, the hypothesis of Proposition 7.5.* In closing the paper (28 September) the Doctoral Examiner stated that a sweep over $\kappa < 0$ violated the conclusions of Proposition 7.5, so that $\kappa \geq 0$ was the right condition and not merely a convenient one. The sweep violated the inequalities of the proof, not the conclusion: the candidate showed that the conclusion holds in every case of the sweep (29 September), and the Doctoral Examiner confirmed it on rederivation the same day. The hypothesis $\kappa \geq 0$ belongs to the proof, and the paper states it as a hypothesis without claiming its necessity for the result.

(m) *Paper 5, Open Problem (v) and Theorem 5.11.* In his first letter of 22 September the Doctoral Examiner wrote that Open Problem (v) of Paper 5 poses as open what the balance law of Paper 6 solves, and asked that it be recorded as solved; his second letter that day withdrew the request as put and suggested a pointer in Open Problem (v) or in the presentation of the portfolio, and his letter of 23 September on Paper 4 (COR-P04-06) described Open Problem (v) as left standing. On 28 September he corrected this: Open Problem (v) asks for a decomposition of the metric charge, and Paper 6 states (Corollary 4.8) that the charge satisfies no additive identity of its own, so the problem is correctly posed as open; Paper 6 proves the decomposition at the deficit level only. The same letter accepted the candidate's restatement of 24 September and corrected a label in his record of 22 September: the residual gap of the joint-law obstruction is the converse of Theorem 5.11(a), not a gap in part (d).

(n) *Paper 6, Open Problem (vi) and Remark 2.7.* In closing the paper (29 September) the Doctoral Examiner described a random search over 114 process-induced families as the more informative test of whether the upper constant is 1. The candidate replied that random maxima understate the supremum, families placed along the worst linear direction at flat families hav-

ing reached 0.999 in his earlier check. The Doctoral Examiner agreed that the supremum is approached near the flat families, where random draws do not land, withdrew the description the same day, and ran the first-order test there. His letter of 21 September contained two further imprecisions, both corrected by the candidate on 24 September: it called the comparison at $d = n$ bi-Lipschitz equivalence, where "comparable" is correct, the statement saying nothing about moduli of continuity, and the Doctoral Examiner accepted the correction; and it cited Example 5.4 as a family carrying laws without transporting them, whereas that family transports its laws and fails to be process-induced for want of a joint law with its three margins. His value $1/2$ for the norm of $LV(2 \leftarrow 1)Q_1$ in Example 4.7 is the size of its entries; its squared norm in the metric of the paper is 1.

(o) *Portfolio-level statements.* The portfolio-level evaluation of 4 September stated that no error was located in any verified mathematical claim across the six works. The later rounds located several, all now corrected and listed in the paper sections and in Section T.9. That evaluation's recommendation against waiving the covering document rested on the misalignment of Paper 1, which the revision has removed (Report, Section 4).

Table T.1: Rounds of the examination, by paper. Dates are 2026, as shown in the Doctoral Examiner's mail (UTC−3). DE: Doctoral Examiner; IE: independent evaluator. The version of record is the text certified in this report.

| **No.** | **First round (text examined)** | **Responses and further rounds** | **Version of record** |
|---|---|---|---|
| 1 | DE report (Findings 1–6, Questions 1–10), dated 4 Sep, sent 11 Sep, on the version of 7 Sep; IE review of 6 Sep, on the earlier version | 7 Sep (to IE); 14 Sep (to DE); 16–18 Sep (IE's two remaining points); DE second round 26 Sep (items 1–5), answered 26 Sep; DE third round 27 Sep, answered 27 Sep; closed 29 Sep, with the corrected DE report | candidate's revision of 27 Sep (14 pp.) |
| 2 | DE report (Findings 1–5, Questions 1–9), sent 7 Sep, on the deposit of 3 Sep; IE review (items 1–10) of 2 Sep, on the text preceding revision 4 | 3 Sep (to IE; revision 4); 21 Sep (IE's comments on the responses); 24 Sep (to both: IE's comments and the DE report; revisions 5–6); DE second round 25 Sep, answered 25 Sep; closed 27 Sep | revision 6.1, 25 Sep (47 pp.) |
| 3 | DE report (Findings 1–5, Questions 1–8), sent 8 Sep; IE review (items 1–29) of 4 Sep, sent again 11 Sep; by the candidate's account both on the same text, then on SSRN | 13 Sep (joint); 17 Sep (IE: no further points); DE second round 23 Sep, answered 24 Sep; DE third round 27 Sep, answered 27 Sep; closed 28 Sep | revision 4, 27 Sep (18 pp.) |
| 4 | DE report (Findings 1–8, Questions 1–16), sent 14 Sep, and IE review (items 1–46) of 11 Sep, by the candidate's account both on the July deposit | 16 Sep (joint); 16 Sep (IE's three points); 18 Sep addendum (revision 2.3); DE second round 23 Sep, answered 24 Sep together with the IE's three points (revision 3); DE third round 28 Sep, answered 28 Sep; closed 28 Sep; 29 Sep (scope of the hypothesis of Proposition 7.5) | revision 3.1, 28 Sep (38 pp.) |
| 5 | DE report (Findings 1–8, Questions 1–18), sent 14 Sep, on the text of 14 July (superseded; Section T.2); IE review (itemised by phrase) of 11 Sep, on the later July deposit | 17 Sep (joint); 18 Sep addendum (revision 2.1); 21 Sep (IE's comments on the responses); DE second round 22 Sep (two letters), answered 24 Sep; closed 28 Sep, with the candidate's undertaking on Paper 6 | revision 2.1, 18 Sep (23 pp.) |
| 6 | DE report (Findings 1–6, Questions 1–16), sent 15 Sep, and IE review of 13 Sep, by the candidate's account both on the August deposit | 18 Sep (joint; revision 2); 21 Sep (IE's comments on the responses); DE second round 21 Sep (five qualifications), answered 24 Sep (revision 2.1); closed 29 Sep; 29 Sep (Open Problem (vi); revision 2.2) | revision 2.2, 29 Sep (23 pp.) |

## T.3 Paper 1 examination: The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization

**Manuscript.** *The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization.* SSRN, DOI 10.2139/ssrn.6922380; earlier version (March 2026) at PhilSci-Archive, ID 28724; titled *The Incompleteness of the Instantaneous State and the Status of Chapman–Kolmogorov Divisibility* in the June 2026 version. Version of record: the candidate's revision of 27 September 2026, 14 pp. (Exhibit B; 01_candidate_submission).

### Technical examiner's verdict

The paper argues along three lines: from the construction of a value at a point as a limit, from Planck-scale and clock bounds, and from renormalization and algebraic field theory. The version of record adds one lemma. Under the Institute's standard for theoretical work the paper is judged on logical consistency and on the testability of what it claims. The version first examined, under its former title, claimed more than its arguments deliver: it drew an ontological conclusion from operational limits, it presented failure of CK composition at fine resolution as a falsifiable signature of an indivisibility-based ontology at a point where the later papers of the portfolio had conceded (Paper 2) or proved (Papers 5 and 6) that ordinary lumping of a Markov chain produces exactly that signature, and it never defined the completeness it argued against. The defects lay in the inference and the framing; the physics cited was correctly described. Each defect was repairable within the paper; together they invalidated the claim of that version. The recommended repair was to state the claim the arguments support, define completeness, and either withdraw the prediction of Section 6 or replace it by something that discriminates.

> **Candidate's response.** The candidate accepted the diagnosis. He records that the paper had been written before the companion papers with a narrower claim, and that later revisions of its title, abstract and discussion had moved it toward a result about CK divisibility that its arguments do not deliver; the revision restores the original claim under the present title. Completeness at a point is now presented as an assumption that the construction of the state does not supply and that established physics gives positive reasons to doubt; Definition 1 separates the deterministic and stochastic senses; CK composition appears as a necessary condition for completeness in the stochastic sense, so that the test of Section 6 is one-sided; and Lemma 1 replaces the prediction. He adopted one recommendation only in part (Section 2 tightened, not cut), with reasons, and contested the grading of the paper's originality. He accepted the grading on the foundational axis in the second round (26 September) and restated the open problem the paper leaves in the third (27 September).

### Findings and responses

#### Finding 6: Dynamical completeness is never defined

The abstract and Sections 6–7 treated the Markov property, CK composition and completeness of the instantaneous state as one condition. They are not: the Markov property implies composition, and the converse fails.

*Severity: major but straightforward.*

**Candidate's response.** Accepted; the candidate identified it as the finding that explains most of the others. Definition 1 (Section 1): in a deterministic theory, a uniquely solvable initial-value problem from data on one slice; in a stochastic theory, the Markov property with respect to the natural filtration. The paragraph after it states the hierarchy (Markov implies composition; Feller's 1959 example shows the converse fails), so that composition is necessary for completeness in the stochastic sense, its failure refutes completeness, and its holding does not establish it. Every inference in Section 6 now runs in the necessary direction (Questions 1–2). Definition 1(a) first said "well posed" and stated no continuous dependence on the data; since the second round it says "uniquely solvable", as does the abstract (Section T.2(f)).

**Doctoral Examiner's disposition.** *Corrected in the version of record; certified against the text.*

### Finding 2: The window claim is false as stated

Section 2 asserted unconditionally that a temporal window makes the observed process non-Markov at every resolution, with a defect vanishing only as the window closes. If the observation is a non-lumpable function of the state, the defect converges to that of the lumped family and no refinement removes it. The examiner's second round supplied a counterexample: a three-state chain whose second and third states are merged, a partition that is not strongly lumpable, with a lumped family whose defect at the times $(0, 0.9, 1.8)$ has spectral norm 0.0024979, independent of $\delta$.

*Severity: major.*

**Candidate's response.** Accepted. A temporal window is a lumping of the path-segment process, not of the state, and the path-segment process of a Markov process is Markov; the Burke–Rosenblatt and Kemeny–Snell results are cited only for the existence of a defect. The refinement behaviour is Lemma 1 (Section 6): for an irreducible finite-state chain started in its stationary law, with maximal jump rate $q_*$, the composition defect of the window average has total variation at most $Cq_*\delta$ for $\delta \le \min\{(t - t_1)/2, 1/(2q_*)\}$, with $C = 2 + 8/c$ explicit; for the window average of a non-lumpable function of the state, the defect converges to the lumped family's, generically positive and independent of $\delta$. The candidate reproduced the examiner's counterexample, which falls in the second branch of the lemma; the paper does not print it, as it reports no computed quantity.

**Doctoral Examiner's disposition.** *Corrected. The examiner re-derived every step of the proof of Lemma 1. His stress tests (four inequalities on 2,000 random generators, with no violation, and a continuous-time simulation) are recorded as tests, not as verification. The paper now states a test that is correctly one-sided.*

### Second round, item 3: The statement of Lemma 1

The proof is correct; the statement did not cover its object. It bounded the defect "for every value $y$ taken by $Y_\delta(t_1)$ with positive probability", and $Y_\delta(t_1)$ has a continuous component whenever a jump falls in the window, with probability of order $q_*\delta$, so the statement controlled the atomic rows only. The bound $Cq_*\delta$ permits $o(\delta)$, and the deposit asserted linear vanishing in three places, the abstract among them. Non-lumpability implies failure of the Markov property but not, by Feller's example, a nonzero two-time defect, so persistence under refinement holds only when the limiting lumped defect is nonzero.

*Severity: material; repairable in standard language without change to the argument.*

> **Candidate's response.** Accepted on all three points. The kernels are the regular conditional distributions of $Y_\delta(u)$ given $Y_\delta(s)$, and the bound holds for almost every $y$ with respect to the law of $Y_\delta(t_1)$; the proof carries over because the bound $\max_x \sum_z |D_x(z)|$ is uniform in $x$. The rate is "at least linearly" in the abstract, Section 2 and Section 6, which applies here the standard of Paper 3, Theorem 5.1, where the rate is stated as an upper rate only. Section 6 adds that the lemma gives no lower bound, so that the defect of a given system may vanish faster than linearly or vanish identically. Section 2 states persistence conditionally on a nonzero limiting defect.

**Doctoral Examiner's disposition.** *Corrected. The examiner had checked numerically that conditioning on a band of the continuous part gives defects of the order of the atomic rows and far below the bound.*

### Finding 1: The apparatus of Section 2

The conclusion of Section 2 follows from the meaning of dynamical sufficiency; in every formalism the paper names, the state at a time is introduced directly and interval averages are derived from it. The section's length outweighs its yield.

*Severity: repairable.*

> **Candidate's response.** Accepted in part. The section now opens by conceding that the conclusion follows from the meaning of sufficiency and that theories introduce point-indexed states directly; the interval average is defined in equation (1) with its convergence condition; and, after the second round, the state at $t$ is the limit of interval averages under a stated observation model, the uniform average of equation (1) being the model used. The section was tightened rather than cut, because the two stipulations the paper identifies — continuity in time and dynamical sufficiency — are located in it.

**Doctoral Examiner's disposition.** *Corrected; the non-adoption of the cut is reasoned and accepted.*

### Finding 3: From spatial to temporal bounds

The inference from the Planck-scale spatial bounds to a temporal one was not supplied.

*Severity: repairable.*

> **Candidate's response.** Accepted. The inference now runs through the position–momentum relation, as the examiner supplied it (localization to $c\,\delta t$ requires momentum $\hbar/(c\,\delta t)$ and energy $\hbar/\delta t$), without invoking a time–energy relation; the clock bound is written $\delta t \gtrsim (T t_P^2)^{1/3}$ in units of time. In response to the second round the candidate scoped the minimum-time language, in nine sentences across the abstract and Sections 1, 3, 5, 6 and 7, to an operational obstruction: five sentences place it within the semiclassical domain of the bounds, four within the domain of established physics.

**Doctoral Examiner's disposition.** *Corrected.*

### Finding 4: Scales conflated in Section 4

The regulator, the renormalization scale $\mu$, the validity scale of an effective description and experimental resolution were run together.

*Severity: repairable.*

> **Candidate's response.** Accepted. The section separates the four; exact observables are independent of $\mu$; the running of $\alpha_s$ is retained only as confirmation of renormalization-group structure. In the second round the regulator became a momentum cutoff $\Lambda_0$ throughout the Wilsonian paragraph, with $\Lambda_0 \to \infty$ for the continuum limit and $\Lambda$ the validity scale only; asymptotic freedom is one route to an interacting continuum limit and an interacting ultraviolet fixed point the other; the bare theory at a point is the $\Lambda_0 \to \infty$ end of the flow; and the abstract no longer identifies the divergences with finite validity.

**Doctoral Examiner's disposition.** *Corrected.*

### Finding 5: Algebraic formulation and Wightman's theorem

The time-slice axiom was not stated, and Wightman's result was attributed in a form ("contradictions under natural assumptions") that does not identify the theorem. In the second round the examiner withdrew the reading of the time-slice axiom the finding had recommended (Section T.2(f)), and asked for Wightman's theorem either for bounded fields or with the domain hypotheses the unbounded case requires.

*Severity: repairable.*

> **Candidate's response.** Accepted. The time-slice axiom is stated (Haag–Schroer; Chilian–Fredenhagen for its proof in perturbatively renormalized scalar theories): the dynamics is determined by a slab of nonzero thickness, however small. The revision of 14 September also concluded, on the reading the finding had recommended, that the slab carries the determination and the surface does not. The version of record ends the sentence at "the axioms assign no algebra to the sharp surface itself" and, in the candidate's words, draws nothing from it. Wightman's theorem was first restated in an unbounded form (vacuum in the domain, $\varphi(x)\Omega = c\Omega$). In the second round the candidate went to the source (Ann. Inst. H. Poincaré A 1 (1964) 403, Théorème 3.1, p. 415), found the theorem stated for fields of bounded operators, with conclusion $B(x) = b\,\mathbf{1}$, and found that the unbounded form was not Wightman's theorem; the paper now states the bounded version.

**Doctoral Examiner's disposition.** *Corrected. The examiner confirmed the bibliographic entry at the source; the statement of the theorem printed in the paper rests on the candidate's reading of the source. The sentence after the time-slice sentence ends with "which is the structure this paper argues for"; the candidate's letter of 26 September describes that sentence as stating the indexing fact and no more. The clause was not raised in the correspondence before the review closed; it is listed as a suggested comment (Report, Section 5.2).*

**Technical report, Sections 3.4–3.6: Measurement, prediction, positioning (Questions 9–10)**

Section 5 claimed more than a dissolution of the collapse-timing formulation; Section 6's prediction did not discriminate; Section 7 lacked positioning against the nearest philosophical work and overstated the domain of the evidence.

*Severity: repairable.*

> **Candidate's response.** Accepted. Section 5 says there is no operationally privileged instant of measurement and that this does not exclude a theoretical jump time; conclusions drawn from the stochastic–quantum correspondence are marked as conclusions inside that framework. Section 6 states a criterion separating classical from quantum memory — Kolmogorov consistency under non-invasive intermediate observation, with the non-invasiveness assumption of Leggett–Garg tests stated — and, after the second round, a sampling floor below which the test is silent. Section 7 positions the paper against Butterfield (2006), Arntzenius (2000) and Albert (2000), and states that the claim is one of warrant, whose empirical stakes are the Planck-scale bound and the refinement behaviour of Lemma 1.

**Doctoral Examiner's disposition.** *Corrected.*

**Question 5 and the verification section**

Question 5 referred to "the two strengthenings of Finding ??" and to a sampling floor, and Finding 2 referred to a counterexample "recorded in the verification section"; neither the finding nor the section was in the report as sent (Section T.2(f)). The second round identified the material. The two strengthenings were the one-sided form of the test and a criterion separating classical memory from quantum indivisibility, both already implemented by way of other findings. The sampling floor was given as a conditional heuristic, $\delta_{\min} \asymp z\,\sigma_{\mathrm{eff}}/(a\,q_*\sqrt{N})$ if the defect behaves as $a\,q_*\delta$ with $a > 0$, to be calibrated per model, since the lemma bounds the defect from above only; with about $2.4 \times 10^5$ trajectories the examiner's sampling error was about $2 \times 10^{-3}$, and at $\delta = 0.01$ the estimated defect of the chain he examined fell below it. He asked that Section 6 require a protocol to declare such a floor, calibrated rather than derived. The counterexample is recorded under Finding 2.

> **Candidate's response.** Section 6 now requires an application of the test to declare a sampling floor, the window width below which the estimated defect is not distinguishable from sampling error at the sample size in hand, calibrated for the model by simulation or bootstrap rather than derived from the lemma; the test is silent below the floor. The heuristic formula for $\delta_{\min}$ is not in the paper.

**Doctoral Examiner's disposition.** *Corrected. The examiner agreed in the third round that the paper is better without a formula it cannot support.*

**Originality (first round; second round, item 5)**

The examiner graded the paper's originality as moderate on the foundational axis and low as new mathematics or physics, and recorded that no prior work assembles these lines or proposes the

criterion. The candidate contested the second axis and asked what a high grade would require of a paper of this kind. In the second round the examiner graded on the foundational axis alone; the corrected report states that the paper is not offered as, and should not be weighed as, new mathematics or new physics. He kept the question of consequences: under the Institute's standard a foundational paper is weighed by whether its thesis does work, measured in consequences, and a higher grade would require a consequence that follows from denying completeness and not from a mis-specified state space, since every positive defect admits the explanation that the state space was badly chosen.

> **Candidate's response.** The candidate accepted the priority finding in the first round, and in the second the grading on the foundational axis and the placement of the paper as the premise of the set. He claims no new physics for the paper and one result of new mathematics, Lemma 1, of modest scope.

**Doctoral Examiner's disposition.** *No correction required. The grade is moderate on the foundational axis, and the paper is the foundational premise of the portfolio, "moderate on its own, load-bearing in the set" (second round). In the third round the examiner recorded that the requirement, posed as a description-independent invariant, is unattainable, and accepted the candidate's restatement of it (below).*

**Independent evaluation (6 September, on the earlier version)**

The candidate's response of 7 September answers the review in twelve numbered points, whose numbers are used here. The independent evaluator found that the paper drew an ontological conclusion from operational limits (point 1); that its abstract overstated what the arguments show and identified generic indivisibility with quantum behaviour (point 2); that the limit argument misread the role of the punctured neighbourhood (point 3); that the Landau–Lifshitz observation could not carry the completeness point, Brownian motion being Markov without a velocity (point 4); that the Planck-scale bounds needed the caveat that they extrapolate both theories beyond their verified domains (point 5); that the Salecker–Wigner paragraph claimed independence from gravitation and then used the Schwarzschild radius (point 6); that the renormalization and algebraic-field-theory statements were overstated or unsourced (points 7–8); that the measurement section claimed too much (point 9); that coarse-graining can create memory as easily as remove it, which bore on the paragraph of Section 6 on $\delta t_{\mathrm{obs}} \gg \tau_{\mathrm{coh}}$ and on its prediction (point 10); and that the closing paragraph of Section 7 contradicted the introduction and the divisibility literature was not cited (point 11). Her comment of 16 September found the changes described in points 1–3 and 5–11 directed at her concerns and raised two remaining points. On point 4, a function of a Markov process is generically, not invariably, non-Markov, and the two-state calculation establishes the behaviour of that example only. On point 12, Sections 3 and 4 may be presented as giving positive reasons to question pointwise completeness provided this remains an interpretative argument, since operational inaccessibility and coincident-point singularities do not by themselves establish dynamical incompleteness; and the burden of justification may be placed on the memoryless description as an argued methodological position, which does not follow from the three physical arguments or the windowed example, its supported form being that memorylessness should not be presumed without testing its stability under changes of temporal resolution and of state-space specification.

*Severity: major (points 1, 6, 10); repairable (the remainder).*

**Candidate's response.** Accepted throughout, with two positions retained (point 12): that established physics gives positive reasons to doubt completeness, resting on Sections 3 and 4; and, as an argued position, that the burden of justification lies on the memoryless description. The quantum mass bound $M \gtrsim \hbar T/(c^2 \delta t^2)$ of the Salecker–Wigner argument now stands alone; the collapse condition and $\ell_{\text{clock}} \lesssim c\,\delta t$ are stated as added assumptions, whose combination gives $\delta t \gtrsim (T t_P^2)^{1/3}$. In checking the bound the candidate found and recorded that its $T$-dependence is disputed (Diósi–Lukács; Baez–Olson; Ng–van Dam). On the two remaining points (18 September): "generically" replaces the universal claim in Sections 2 and 6, the two-state calculation is not in the paper, and Lemma 1 gives an $O(\delta)$ bound for a class; Section 3 says that the operational result does not establish incompleteness of the mathematical state, and Section 4 that the argument is evidential, not deductive; and Section 7 presents the reversal of the burden of proof as the conclusion argued, in her operational form, with the two halves of Lemma 1 as its two tests.

**Doctoral Examiner's disposition.** *Certified against the version of record. The independent evaluator stated (16 September) that she had assessed the candidate's descriptions of the changes, not the revised text, and that confirming their implementation would require a separate review; the Doctoral Examiner has certified the text.*

### Second round (26 September, items 1–5)

The examiner's assessment was favourable: both material findings (Findings 2 and 6) were resolved in the direction both reports had asked for. Item 1 corrected his first-round report, withdrew two prescriptions (Section T.2(f)) and supplied the missing material (Question 5). Item 2 recorded what is resolved, including two additions to the text: a sentence neither report had asked for, stating that the presupposition concerns the completeness of the state at a point, "not its existence, which is a matter of definition and is not in dispute here", which answers the independent evaluator's principal objection; and the observation that, under the first hypothesis, the limit point has a property its approximants generically lack. Item 3 concerned the statement of Lemma 1 (above); item 4 asked for four amendments (Findings 1, 3, 4 and 5); item 5 took up the grading of originality and the missing invariant.

**Candidate's response.** The revised deposit of 26 September carries every amendment of items 1, 3 and 4; "nothing in item 2 changed". The responses to item 5 are recorded under "Originality" and in the next finding.

**Doctoral Examiner's disposition.** *Corrected; certified against the version of record.*

### The open problem of admissible descriptions (second round, item 5)

The examiner wrote that a higher grade would require a description-independent invariant, whose positive value cannot be absorbed by re-specifying the state space. He withdrew a statement of his working record that the later papers supply it: the deficit with its locus and $\beta$, the liftability profile and the memory rank are invariants relative to a transition family and an observed space already fixed, and Paper 2's Remark 2.8 records that unrestricted history augmentation always Markovizes (Section T.2(g)). He identified the gap as the principal open problem of the portfolio as a whole, not a defect of Paper 1.

**Candidate's response.** The candidate agreed that the invariant is missing and showed that it cannot exist in the form asked: augmenting the state with its full history renders any process Markov, so no invariant independent of every description exists. An invariant can exist relative to a physically fixed class of admissible descriptions, and the open problem is that class. The paper fixes it only in part, by the resolution bound of Section 3 and the non-invasiveness condition of Section 6, and Section 7 says so. After the third round he named the missing constraint as a bound on history depth or state-space dimension, the quantity the memory rank of Paper 5 measures relative to a fixed description, and posed as a question, without claiming an answer, whether that quantity can be made invariant within a class bounded in dimension.

**Doctoral Examiner's disposition.** *Accepted by the examiner in the third round, with the argument written out: for $\tilde{X}_t = (X_s)_{s \le t}$ one has $\sigma(\tilde{X}_s) = \mathcal{F}_s$, so the path-space description is Markov for every process and its defect vanishes identically, and a quantity invariant under change of description and nonzero somewhere would have to vanish there. The examiner recorded that his own formulation had stated the problem in an unattainable form. In the closing letter (29 September) he gave the scope of the candidate's question: bounding the dimension excludes the path-space description, which blocks the impossibility argument, but the memory rank is defined relative to a fixed observed state space and transition family; under a change of description the family changes and the rank with it, so bounding the class does not by itself make the rank invariant. The open problem is the portfolio's; Section 7 of Paper 1 states it (Report, Section 3.1).*

### Third round (27 September) and closure (29 September)

The examiner confirmed the implementation of items 1, 3 and 4: the lemma on regular conditional distributions, with the uniformity in $x$ carrying the bound; "at least linearly" in all three places, with the added statement that the lemma gives no lower bound, which he judged worth more than the amendment he had asked for; persistence conditional; the sampling floor declared and calibrated, without the heuristic formula; $\Lambda_0$ and $\Lambda$ separated; the time-slice sentence, which "stops where it should"; "uniquely solvable"; consistent modal language; and Wightman's theorem taken from the source. He accepted the candidate's correction of the open problem and added one statement: neither constraint the paper supplies acts where the impossibility bites. The resolution bound restricts operational access in time and the non-invasiveness condition restricts the observation protocol; neither bounds history depth or state-space dimension, and a path-space state on a coarse time grid respects the Planck bound, is non-invasive and has zero defect.

**Candidate's response.** The candidate added the statement as the last sentence of Section 7 and changed nothing else in the deposit of 27 September, the version of record: "Neither constraint bounds history depth or state-space dimension: a history state on a time grid coarser than the resolution bound satisfies both conditions and has zero defect, so the class is unpinned where augmentation acts."

**Doctoral Examiner's disposition.** *Closed with no outstanding amendments (29 September). The examiner audited the candidate's declaration that nothing else changed. The corrected technical report, TR-P01-DOCv2-EX01-Rv2, was sent with the closing letter; it "adds no correction to Revision 1 (the four amendments there stand)", and its object remains the deposit of 7 September (Exhibit I).*

## Further corrections identified by the candidate

> **Candidate's response.** The candidate made three corrections beyond what the findings asked: the finding, at the source, that the unbounded form of Wightman's theorem printed on 14 September was not Wightman's theorem; the disputed $T$-dependence of the clock bound; and the reformulation of the open problem relative to a physically fixed class of admissible descriptions.

**Doctoral Examiner's disposition.** *The self-identified corrections are in the version of record. Defects disclosed by the candidate beyond the examination's findings are weighed as evidence under the governing standard.*

## Corrected architecture adopted

1. Definition 1 of dynamical completeness in two senses, and the hierarchy Markov $\Rightarrow$ composition $\not\Rightarrow$ Markov (Section 1).
2. The limit: continuity in time and dynamical sufficiency as the two stipulations, under a stated observation model (Section 2).
3. Planck-scale and clock bounds as an obstruction within the semiclassical domain (Section 3).
4. Renormalization scales separated; localization of observables and the time-slice axiom (Section 4).
5. The collapse-timing formulation of the measurement problem (Section 5).
6. Lemma 1 for almost every conditioning value, the one-sided test, the declared sampling floor, and the consistency criterion for classical versus quantum memory (Section 6).
7. Positioning; the open problem of admissible descriptions, with the statement that neither constraint of the paper bounds history depth or state-space dimension (Section 7).

## Classification of results

*Sound and substantial:* the foundational argument as now scoped (on the foundational axis); Lemma 1, correct and of modest scope. *Requiring correction:* none outstanding (the clause noted under Finding 5 is a suggested comment). *Incorrect as stated (earlier versions; withdrawn or corrected):* the claim about CK divisibility; the prediction of Section 6; the unconditional window claim; the independence claim of the clock argument; the inference from the time-slice axiom to the sharp surface; the unbounded form of Wightman's theorem; the statement of Lemma 1 for the atomic values only.

## Correspondence (Exhibit A)

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P01-01 | 6 Sep | EX02 → CAN | Independent evaluation transmitted, as forwarded by the candidate to EX01 on 7 Sep; includes her message of 4 Sep transmitting her evaluation of Paper 3 |
| COR-P01-02 | 7 Sep | CAN → EX02, cc EX01 | Response in twelve points; revision of 7 Sep attached |
| COR-P01-03 | 11 Sep | EX01 → CAN | Technical report transmitted |
| COR-P01-04 | 14 Sep | CAN → EX01 | Response to Findings 1–6 and Questions 1–10 (Question 5 deferred); three defects of the report; the originality grade contested; revised deposit |
| COR-P01-05 | 16 Sep | EX02 → CAN, cc EX01 | Two remaining points (4 and 12) |
| COR-P01-06 | 18 Sep | CAN → EX02, cc EX01 | Answer to points 4 and 12; revised deposit |
| COR-P01-07 | 26 Sep | EX01 → CAN | Second round, items 1–5 |
| COR-P01-08 | 26 Sep | CAN → EX01 | Response to the second round; revised deposit |
| COR-P01-09 | 27 Sep | EX01 → CAN | Third round: the correction of the open problem accepted; one addition |
| COR-P01-10 | 27 Sep | CAN → EX01 | The addition ends Section 7; version of record deposited |
| COR-P01-11 | 29 Sep | EX01 → CAN | Closing letter; corrected report TR-P01-DOCv2-EX01-Rv2; Paper 1 closed |

## T.4 Paper 2 examination: Koopman Spectral Theory for Indivisible Stochastic Processes

**Manuscript.** *Koopman Spectral Theory for Indivisible Stochastic Processes.* SSRN, DOI 10.2139/ssrn.6733181. Version of record: revision 6.1, 25 September 2026, 47 pp. (Exhibit C; 01_candidate_submission).

### Technical examiner's verdict

The paper is the technical core of the programme and its strongest paper. Its mathematics is competent, honest and carefully bounded, with a new invariant at its centre. The framework works on the forward family $\{P(t,s)\}_{s\le t}$ through the singular structure of the deficit, correctly justified (two-time operators are generically non-normal, and compactness gives no eigenexpansion). The sound core is Proposition 4.12 and the indivisibility tensor $\beta(s) = C(s) - L(s)^2 - L'(s)$, verified on a construction with $\beta$ in closed form, genuine $s$-dependence and non-commuting kernels; Theorem 4.8 with its counterexamples; and the global deficit locus with the three-pair obstruction. The defects lay in proportion between claim and content (several statements are immediate or classical), priority (the identity behind $\beta$, earlier dilation constructions), scope (a generator statement stated without its class), and errors of statement or proof found by the independent evaluator (items 1, 2, 7, 8 and 9). No defect was fatal to a result on which the paper's contribution rests; one rendered Theorem 7.4 vacuous in infinite dimensions until repaired. The recommended repair was to label standard material, attribute, scope, and settle the open optimality question of Remark 4.7.

> **Candidate's response.** The candidate adjudicated each point of both reviews individually and reports that revision 5 implements every item except a change of title, which he declined with a reason now stated in the introduction, and Question 7, which he answered in correspondence rather than in the paper. He answered the independent evaluation on 3 September (revision 4), and her comments of 21 September together with the technical report on 24 September (revisions 5 and 6); revision 6.1 (25 September) made the two amendments of the second round. He confirmed the examiner's optimality finding and strengthened it to an exact result (Proposition 4.7), and added a product rule giving closed-form families in every dimension $2^k$ (Section 6.8). His letters on Papers 3–6 record that every result they cite from this paper keeps its number.

### Findings and responses

#### Finding 4 and Question 2: Optimality of the constant 1/3

Remark 4.7 left open whether the constant 1/3 in the three-pair obstruction is optimal. The examiner showed that it is optimal, by a first-order argument in the two-state symmetric reduction, with the family $\Gamma(T, T/2) = \Gamma(T/2, 0) = I$, $\Gamma(T, 0) = I - \frac{D}{2}(I - X)$.

> **Candidate's response.** Confirmed and extended (Proposition 4.7): every divisible column-stochastic comparison has $\sup \delta \ge m(D) = (3 - \sqrt{9 - 4D})/2$, attained by an explicit semigroup, with $m(D)/(D/3) = 1 + m(D)^2/D \to 1$; the lower bound holds for all column-stochastic comparisons, by a rank-one reduction; the family is extended to continuous time so the statement holds on the whole simplex; and the bound is strict for every $D > 0$, approached and not attained.

**Doctoral Examiner's disposition.** *Correct. Every step of the rank-one reduction checks. The examiner confirmed independently the global minimality of $m(D)$ over the full four-parameter class for $D \in [0.2, 1.9]$, the attainment by the semigroup with the supremum taken over all pairs and not only the three of the triple, and the monotonicity step of part (b), numerically up to $D = 1.99$; the version of record proves that step for $0 < D < 2$. The proposition closes the former open remark. The examiner's own wording, which implied attainability, is corrected (Section T.2(e)).*

### Finding 1 and Question 1: Priority for the identity behind beta

The identity $C = A' + A^2$ for $\partial_t U = AU$ is classical (Dyson, Magnus); the paper contributes the constructions built on it.

*Severity: expositional (attribution).*

> **Candidate's response.** Accepted. Dyson and Blanes–Casas–Oteo–Ros are cited, followed by a sentence stating what the section adds: the residual as a named invariant, the factor $(t - r)(r - s)$, the singular-direction reading, and the place of $\beta$ in the global theory.

**Doctoral Examiner's disposition.** *Corrected.*

### Finding 5 and Question 4: Priority for the dilation

Earlier constructions (Canturk; Doukas) provide the basic Stinespring lift of a stochastic family.

*Severity: expositional (attribution).*

> **Candidate's response.** Accepted. Both are cited, and Section 5 no longer claims the basic lift. Section 5 still claims the diagonal-retention channel with $\Phi(s, s) = \mathrm{id}$, the jointly continuous completion with $W(s, s) = I$ on a fixed environment of dimension $n^2 - n + 1$, the formal generator under (R3′), the grid cocycle, and the incorporation into the deficit framework.

**Doctoral Examiner's disposition.** *Corrected.*

### Finding 2 and Question 5: The formal Hamiltonian is gauge-dependent

The formal generator $\hat{H}$ of the dilation depends on the choice of continuous dilation; the paper did not say so.

*Severity: repairable.*

> **Candidate's response.** Accepted. Remark 5.7 states that a formal Hamiltonian generator can be constructed for a selected continuous dilation and gives the gauge law: under $W \mapsto (I \otimes U(t))W$ the channel is unchanged and $\hat{H}$ transforms by conjugation plus $i\dot{U}U^*$, so no feature of $\hat{H}$ is invariant; the invariants are the channel family, the populations and the deficit.

**Doctoral Examiner's disposition.** *Corrected. Since $\dot{U}U^*$ ranges over all anti-Hermitian functions, the conclusion is exact.*

### Proportion and priority

Theorem 4.5(a) is immediate from $\ker \Delta = \operatorname{ran}(\Delta^*)^\perp$; Corollary 4.6 is a triangle inequality; Theorem 5.4 is largely an assembly of Stinespring, Kretschmann–Schlingemann–Werner, Kato and Steenrod; the sewing and almost-flow literature is not acknowledged.

*Severity: expositional.*

> **Candidate's response.** Accepted. Proposition 3.6 carries the scope-fixing label used for Proposition 2.6; Theorem 4.5(a) is marked as a structural consequence of the subspace definitions and the Schmidt decomposition; Corollary 4.6 is described as a triangle inequality whose content is the constant; Theorem 5.4 has a lead-in separating the imported machinery from the diagonal-retention indexing; a related-work paragraph acknowledges sewing and almost-flows, and Table 1 gives a theorem-level comparison with the three nearest programmes (Questions 3, 6).

**Doctoral Examiner's disposition.** *Corrected. No broad priority language remains.*

### Questions 3 and 6–9

The questions concerned the nearest notions to the deficit subspaces (Question 3), the collision-model structure of the grid dilation (Question 6), an estimator for $\beta$ (Question 7), what a measured deficit establishes (Question 8), and non-vacuity beyond two and three states (Question 9).

> **Candidate's response.** Answered. Question 3, in correspondence: the nearest notions are the Sz.-Nagy–Foiaş defect spaces of a single contraction, the sewing defect of almost-flows and Tian's defect kernel of a compressed semigroup (June 2026); none spans the ranges of all composition deficits over all triples, and that global locus, with vanishing on $V_{\mathrm{in}}^{\perp}$ and the obstruction of Theorem 4.5(b), is the claim the candidate defends, the pointwise range and kernel facts being linear algebra. Question 6: Theorem 5.4(iii) is a collision model with a fresh environment factor per grid increment; the distinctions from the process-tensor framework are classical input, a fixed environment per pair and joint continuity in $(t, s)$, and Table 1 lists the counterparts. Question 7 is answered in correspondence with a concrete protocol (estimate $\beta$ from the slope of $\hat{\Delta}$ against $(t-r)(r-s)$; relative precision $\rho$ needs $N \gtrsim (\rho h^2 \|\beta\|)^{-2}$ transitions per pair) and kept open in the paper. Question 8: a measured deficit establishes non-divisibility of the chosen description and the absence of a Dyson-class generator for it, not intrinsic memory (Remark 2.8, Section 8); the letter adds that any finite-memory process is a lumping of a Markov process on an augmented space and that a compressed semigroup has a nonzero defect. Question 9: new Section 6.8, with a product rule for $\Delta$ and $\beta$ (Proposition 6.1) yielding closed-form families in dimension $2^k$.

**Doctoral Examiner's disposition.** *Answered; Section 6.8 verified (diagonal form, closed-form norm, the singular values $4\varepsilon, 2\varepsilon, 2\varepsilon, 0$ of $\beta_\otimes$, rank and mean-zero statements, the $2^k$ iteration, agreement with the scaling law near the diagonal). The estimator and its bootstrap theory are undeveloped, so leaving the estimator as an open problem is correct. The nearest notions named for Question 3, other than the sewing literature, appear in the correspondence only. In the second round the examiner withdrew an objection to the phrase "leading direction $v \otimes v$" in the candidate's letter of 24 September: the leading singular direction of the product deficit is $v \otimes v$ only when $\alpha(t,s) + \alpha(t,r)\alpha(r,s) > 1$, and Section 6.8 states that criterion, $D_\otimes = |\delta| \max\{1, \alpha(t,s) + \alpha(t,r)\alpha(r,s)\}$, and attaches the leading direction to $\beta$. The candidate agreed (25 September) that the sentence of his letter was loose.*

**Independent evaluation (2 September; comments of 21 September)**

The independent evaluator's review (2 September), of the text preceding revision 4, found: the witness $\frac{1}{2}(J_n - I_n)$ of Proposition 2.11(c) is doubly stochastic only for $n = 3$, so the clause failed for every $n > 3$ (item 1); the augmentation passage of Remark 2.8 was false as written, history augmentation Markovizing any process (item 2); Section 3.2 asserted semigroup structure before separating time-homogeneous from evolution families (item 3); the generator statement lacked its class (item 4); the text after Corollary 4.6 called the bound invariant under the choice of $r$, on which $D(t, s, r)$ depends (item 5); Michael's selection theorem was applied in Appendix C, Step 1b, to a correspondence that is not convex-valued (item 7); minimality in Theorem 5.4(iv) holds only per pair $(t, s)$ (item 8); and hypothesis (C2) of Theorem 7.4 — norm continuity on the closed simplex — together with compactness for $t > s$ and $P(s, s) = I$ would make the identity a norm limit of compact operators, impossible in infinite dimensions (item 9). Her item 6 read hypothesis (R3′) as placed on $\Gamma_{ij}$ rather than on $\sqrt{\Gamma_{ij}}$. Item 10 was editorial: the length of the abstract, the classical status of Proposition 2.6 and Corollary 2.7, what "Koopman" does in the title, caveats on the empirical protocol, the collision-model structure of the grid dilation, the positioning of Theorem 4.5(a), and a theorem-level comparison with non-Markovian transfer-operator, cocycle-defect and process-tensor approaches. Her comments of 21 September addressed the candidate's explanations only, not revision 4 or the revised proofs. She withdrew item 6, found the other corrections appropriately directed, and noted that removing the Michael argument does not by itself verify the completion argument along the diagonal. She asked that two qualifications be made explicit: the contraction assumption at every statement of the 1/3 bound, and the operational restrictions and the Bell-type results behind the augmentation passage, named. She asked for the theorem-level comparison, which explaining the title and citing the collision-model review did not supply. On Theorem 4.5(a) she was content with the existing remark if it identified the Schmidt machinery, the useful distinction being between the structural consequence of the subspace definitions and the dynamical information established afterwards.

*Severity: major for items 7 and 9 (a proof step and a hypothesis rendering a theorem vacuous), repairable for items 1–5 and 8, expositional for item 10.*

**Candidate's response.** Accepted on 3 September (revision 4), except item 6. Proposition 2.11(c) gives the padded witness $\frac{1}{2}(J_3 - I_3) \oplus I_{n-3}$ with a block-diagonalization proof. Remark 2.8 distinguishes unrestricted augmentation, always available, from augmentation respecting the operational constraints of the correspondence setting. Section 3.2 separates elapsed-time semigroups from evolution families; the generator statement is confined to the strongly measurable, Bochner-integrable class; the bound after Corollary 4.6 holds for every $r$, its strongest form being the supremum over $r$, with a qualifier that it concerns uniform operator-norm approximation of the two-time family. The Michael clause is deleted, Steenrod triviality being the primary route and Kato the alternative; minimality in Theorem 5.4(iv) is per pair; (C2) is restated on the open region, which the proof never left, and Remark 7.5 gives the argument. Item 10 was accepted as a package (the abstract cut, the classical background labelled, the title explained, the empirical caveats added with Davis–Kahan and Wedin, the collision-model review cited), except that the remark before Theorem 4.5 was kept; the theorem-level comparison followed on 24 September. On item 6 the candidate showed that the hypothesis had been on the square roots verbatim since the first revision, and added a parenthetical stating why. On 24 September (revision 5) he implemented what he counted as four open points: the contraction assumption at every statement of the 1/3 bound, including the lead-in to Corollary 4.6 and the Discussion summary; Bell locality, excluded by Bell's theorem in Fine's form, and noncontextuality, excluded by Kochen–Specker in dimension at least three, named and cited, with no general non-augmentability claimed; Theorem 4.5(a) marked as a structural consequence of the subspace definitions and the Schmidt decomposition, the

dynamical content being located in part (b), Corollary 4.6, Proposition 4.12 and Theorem 7.4; and Table 1, which compares the three programmes theorem by theorem. On the diagonal: by (R1), $P(s,s)$ is the projection onto $H\otimes|0\rangle$, so $I-P(t,s)$ has constant rank on the closed simplex; Steenrod triviality over the contractible base gives a continuous frame of its range, and rotating that frame by $G(s)^{-1}$, which depends on $s$ alone, gives $W(s,s)=I$.

**Doctoral Examiner's disposition.** *Corrected; certified against the version of record. The examiner re-examined the completion argument in the second round: the rotation by $G(s)^{-1}$ supplies the relative trivialization the earlier argument lacked, preserves joint continuity in $(t,s)$, and is valid in the finite-dimensional setting in which it is used. He confirmed the contraction qualification at all four locations, the attributions of the augmentation passage, and Table 1 as answering the evaluator's request in the register she specified. The examination's own audit had missed item 7, and the portfolio-level evaluation misattributed items 1, 8 and 9 to the candidate (Section T.2(c),(d)).*

### Second round (25 September) and closure (27 September)

The examiner re-derived every new claim against the deposited text and found the revision favourable: the mathematical, interpretive and priority issues of both reports had in substance been addressed. He asked for two amendments: Table 1 said that the objects of Theorem 4.8 vanish under the cocycle, whereas only their invariant sum does, as Theorem 4.8(c) and Remark 4.9 show; and the Discussion said the constant is approached "only by" families with deterministic intermediate kernels, which Proposition 4.7 does not prove and which, in any quantitative form, is equivalent to the subclass question the same sentence declares open.

**Candidate's response.** Both made in revision 6.1 (25 September), with nothing else changed: Table 1 lists Theorem 4.8 separately: the invariant sum vanishes under the cocycle, and the gauge-dependent summands need not; and the clause now reads "approached by the extremal families exhibited in Proposition 4.7, whose intermediate kernels are deterministic". The candidate has no rigidity argument, and the subclass question stays open as the statement of what is missing.

**Doctoral Examiner's disposition.** *Corrected. The examiner verified both amendments by differencing the two deposits (27 September): the only substantive changes are the two required. He audited the referent of every entry in the first row of Table 1, a step that, by his letter of 27 September, he had skipped in the second round: under the cocycle $\Delta\equiv 0$, so $V_{\mathrm{in}}=V_{\mathrm{out}}=\{0\}$ and the obstruction of Theorem 4.5 is trivial, Corollary 4.6 bounds a vanishing quantity, $\beta$ vanishes and $J(R)=0$ by Theorem 7.4, and Theorem 4.8 is the only one of the five results in that row whose content does not vanish. Revision 6.1 is the operative text, identifiable by its date pair and by Proposition 4.7 and Table 1. The review was closed with nothing outstanding of content or of record.*

### Further corrections identified by the candidate

**Candidate's response.** In answering the findings the candidate went past what was asked: he established the strictness of the bound of Proposition 4.7 (Finding 4), which corrected the examiner's wording; gave the relative trivialization of the completion argument (the evaluator's comment of 21 September); and added the product rule of Section 6.8 (Question 9).

**Doctoral Examiner's disposition.** *These corrections are in the version of record and are weighed as evidence under the governing standard.*

## Corrected architecture adopted

1. Two-time families and scope-fixing classical background (Section 2), including the corrected Proposition 2.11(c) and Remark 2.8.
2. Divisible families in two regimes; no generator in the Dyson (Bochner-integrable) class (Section 3).
3. Deficit subspaces, the three-pair obstruction under contraction, its sharpness (Proposition 4.7), the gauge-relative decomposition (Theorem 4.8) and the indivisibility tensor (Proposition 4.12) (Section 4).
4. The diagonal-retention dilation and its gauge law (Section 5).
5. Worked families, including $V_{\text{in}} \neq V_{\text{out}}$, and the product rule (Section 6).
6. The integrated deficit on the open region (Section 7); the empirical protocol with its caveats and the open problems (Section 8).

## Classification of results

*Sound and substantial:* Proposition 4.12 and the tensor $\beta$; Theorem 4.8 with Remarks 4.9 and 2.12; Theorem 4.5(b), Corollary 4.6 and Proposition 4.7; Theorem 5.4 (diagonal-retention indexing); Theorem 7.4 on the open region; Section 6. *Requiring correction:* none outstanding. *Incorrect as stated (earlier versions; corrected):* Proposition 2.11(c) for $n > 3$; hypothesis (C2) on the closed simplex; the appeal to Michael's theorem; the augmentation claim of Remark 2.8.

## Correspondence (Exhibit A)

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P02-01 | 2 Sep | EX02 → CAN | Independent evaluation transmitted, as forwarded by the candidate to EX01 on 2 Sep; includes earlier messages between her and the candidate of 27–28 Aug on her engagement (Report, Section 6) |
| COR-P02-02 | 3 Sep | CAN → EX02, cc EX01 | Response to items 1–10; revision 4 attached |
| COR-P02-03 | 7 Sep | EX01 → CAN | Technical report transmitted, on the version then on SSRN (the deposit of 3 Sep) |
| COR-P02-04 | 21 Sep | EX02 → CAN | Comments on the candidate's responses for Papers 2, 5 and 6, as forwarded to EX01 on 24 Sep |
| COR-P02-05 | 24 Sep | CAN → EXAMINERS | Answer to the evaluator's comments and to the technical report; revisions 5 and 6 |
| COR-P02-06 | 25 Sep | EX01 → CAN, cc EX02 | Second round: two amendments; correction of Finding 4; one objection withdrawn |

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P02-07 | 25 Sep | CAN → EX01, cc EX02 | Revision 6.1, with the two amendments |
| COR-P02-08 | 27 Sep | EX01 → CAN, cc EX02 | Amendments verified by differencing the deposits; review closed |

## T.5 Paper 3 examination: Convex Perturbation and the Suppression of the Indivisibility Deficit

**Manuscript.** *Convex Perturbation and the Suppression of the Indivisibility Deficit.* SSRN, DOI 10.2139/ssrn.6812461. Version of record: revision 4, 27 September 2026, 18 pp. (Exhibit D; 01_candidate_submission).

### Technical examiner's verdict

The paper is correct, well written, honest and incremental, and a sound minor member of the programme. Theorem 3.1 is the exact identity $\Delta(t, s, r; \Gamma_g) = (1 - g)[X - gK]$. Because the identity is exact, the non-monotonicity of the response in the coupling is an algebraic fact rather than an estimation artefact, and that fact forces the two-threshold formulation: a first crossing can be followed by re-entry, so only a terminal threshold supports the operational reading. The worked example and its rebound were reproduced exactly. The depth is modest (a bilinear expansion, the triangle inequality, convexity, compactness). The defects concerned the status of the model (the mixture was presented as the canonical model of a state-conditioned interaction, and the discussion drew the interpretive conclusions the abstract disclaimed), a false uniqueness inference in Section 7.1, and the structure of the threshold bounds. Both reviews were prepared without conferring and, by the candidate's account, on the same text; they converged on these three. None was fatal. The recommended repair was to present the mixture as a formal path, correct the inference, and state the bounds at their true strength.

> **Candidate's response.** The candidate accepted all three and added one result of his own, Proposition 3.2: the deficit up to scale $\delta t$ is exactly $(1-g)S(g)$ with $S(g) = \sup\|X - gK\|$ convex and Lipschitz, which locates exactly when the mixed family is divisible at an interior coupling. He answered both reviews in one letter (13 September), adopted the two further results and four wording points of the second round (24 September), and made the amendment and the rider of the third (27 September). He has no counterargument to the assessment that the depth is modest, and prefers a short correct paper to a longer one.

### Findings and responses

#### Finding 1: Verification record; the two-threshold structure

The examiner's verification reproduced the identity on random instances and the worked example exactly, and endorsed the two-threshold structure as the paper's methodological contribution.

> **Candidate's response.** The candidate would defend the two-threshold point; it stands unchanged.

**Doctoral Examiner's disposition.** *No correction required.*

**Finding 2 and Question 3: A false uniqueness inference in Section 7.1**

Section 7.1 stated that exact monotonicity is restored "that is, at $g = 1$", inferring that exact vanishing occurs only there. The paper's own interaction-dominant example has an interior zero, at $g = 0.3603$.

*Severity: major but straightforward.*

> **Candidate's response.** Accepted. The clause had been introduced in the July revision, in correcting an earlier overclaim, after Buscemi–Datta, that the terminal threshold restores information monotonicity; the correction replaced the overclaim with a false inference. The correct statement is Proposition 3.2(iii): for $g < 1$ the mixed family is divisible up to scale $\delta t$ if and only if $X = gK$ at every triple, so an indivisible substrate has at most one interior divisible coupling, $\|X\|/\|K\|$; in the example $K = cX$ with $c \approx 2.78$ and the interior zero is $1/c = 0.3603$. Above the terminal threshold the deficit is suppressed to tolerance, not reduced to zero.

**Doctoral Examiner's disposition.** *Corrected; Proposition 3.2 checked. In the second round the examiner recorded that Proposition 3.2 replaces the false inference by a characterization, which is more than the correction required.*

**Finding 3 and Question 1: What the convex combination models**

No dynamical mechanism yields a convex mixture of transition kernels at fixed weight: a random choice between two dynamics, made once and conditioned later on the state, gives a weight that Bayes' rule updates with the state and the time.

*Severity: major (status of the model).*

> **Candidate's response.** Accepted. The mixture is now the simplest fixed-weight convex perturbation of an indivisible family, a formal path in the convex set of two-time families; "canonical" and "observer coupling" are removed, and $g$ is the mixing weight. Remark 2.7 states what the paper does not claim. Remark 6.2 computes the branch-mixture weight $g\pi_s^Y(j)/[g\pi_s^Y(j)+(1-g)\pi_s^Z(j)]$, which for $0 < g < 1$ is constant only when the one-time laws coincide, so the constant-weight model is the marginal-matched special case and Proposition 6.1 covers the general branch mixture.

**Doctoral Examiner's disposition.** *Corrected (restated). In the second round the examiner recorded that Remark 6.2 answers the finding by computing the weight rather than withdrawing the claim, the computation being the substance of the objection.*

**Finding 4 and Questions 4, 6: Interpretation in Section 7.5**

The abstract declared "observer coupling" a label without interpretive weight; Section 7.5 drew interpretive conclusions from it, and detectability was stated without the statistical apparatus it requires.

*Severity: repairable.*

**Candidate's response.** Accepted. Section 7.5 is a scope statement: detectability is deterministic and population-level; from data, "below $\varepsilon$" requires an estimator, its null distribution and a calibration, which the paper does not supply and says so; "indistinguishable" is qualified throughout; the deficit subspaces of the mixed family move with $g$.

**Doctoral Examiner's disposition.** *Corrected.*

### Finding 5 and Question 2: Priority

The terminal threshold is a fixed-reference, tolerance-relaxed relative of the Markovian-admixture measures of Anand–Brun and De Santis–Giovannetti; the examiner set out four distinctions. Non-additivity under mixing (Xu–Jagadish) was raised alongside.

*Severity: expositional (attribution).*

**Candidate's response.** Accepted. All three are cited; the introduction and a new Section 7.3 state the four distinctions and place the originality in the setting and the identity, not in the idea of Markovian admixture.

**Doctoral Examiner's disposition.** *Corrected.*

### Questions 5, 7, 8

The questions concerned the no-generator statement (Question 5), the invariant meaning of the interaction scale $M_*$ (Question 7), and the rate form of the stability result (Question 8).

**Candidate's response.** The no-generator statement is scoped to the Bochner class and appears once. $M_*$ has no invariant meaning beyond being $\sup\|K\|$; the invariant object is $S(g)$, and the bounds summarize the pencil $X - gK$ by two numbers, which discards sign by construction (Remark 4.5). The rate form is withdrawn (Section 7.4).

**Doctoral Examiner's disposition.** *Answered and corrected. Question 7 was settled by the attainment condition of Remark 4.5: the second round gave it a pointwise witness and the third a scale-level witness, Proposition 4.10(ii) (below).*

### Independent evaluation (4 September, items 1–29)

The independent evaluator found, among other points: an abstract claiming "arbitrarily strong suppression at a computable sub-unit coupling" (items 1–2); an unproved specialization to process tensors, contradicted by Section 7.2, and overstated transfers to non-Markovianity functionals (items 5–11); conventions and orders stated in the wrong variable (items 12–17); strict inequalities in the threshold definitions, and a quadratic envelope sharper than the linear bound (items 18–19); a remark on mixing estimates whose gap-only bound requires a reversible or normal generator

(item 20); a rate scale with no independent content (items 21, 28); a single-triple example presented as scale-level (item 22); a Minkowski form of the integrated bound (item 23); and the observer-coupling inconsistency (item 29). Her review was sent to the candidate on 4 September and again on 11 September.

*Severity: repairable; item 20 (a remark resting on false hypotheses) major for that remark.*

> **Candidate's response.** Accepted, with two decisions about length: the example was not at first extended to a full family (since superseded by Proposition 4.10), and the retarded-operator material of Section 5 was compressed rather than removed. The quadratic envelope is now the main statement of Theorem 4.3, $g_{\mathrm{term}} \le g_1 = 1 - [D + M - \sqrt{(D+M)^2 - 4M\varepsilon}]/(2M)$, with the linear bound as a corollary; the mixing remark and its references are deleted; Theorem 5.1 takes the Minkowski form.

**Doctoral Examiner's disposition.** *Corrected. The evaluator replied on 17 September that the changes the candidate described address the substance of her comments, that both non-adopted recommendations are reasonable (the pointwise example, provided its status and its distinction from the scale-level results are explicit; Section 5 in compressed form, for its connection with the integrated deficit of Paper 2), and that she had no further points arising from his response "at this stage". She was copied on the later rounds and made no further comment.*

### Second round (23 September): attainment and a scale-level family

The examiner found the three principal issues addressed. His verification covered the identity (6) on 4,000 random instances in dimensions 2 to 6; Proposition 3.2; Theorem 4.3 on 3,000 random parameter sets; the constant $\eta(\varepsilon)$ of Proposition 4.8; the Minkowski form on 300 families; Proposition 6.1 on 1,500 random weights; the Bayes weight of Remark 6.2 on 2,000 kernel-and-law pairs; every number in Section 4.3; and the internal numbering after the deletions. He recorded that the candidate had carried the quadratic-envelope device over to Proposition 4.8 with the discriminant $(m_0 - M)^2 + 4M\varepsilon$. He proposed two results and supplied their constructions, neither a correction and neither a condition of his recommendation: an attainment condition showing that no bound in $D_{\mathrm{CK}}$ and $M_*$ alone improves Theorem 4.3, with a realization at $\lambda = 0.35$ in the example of Section 4.3; and a scale-level family exhibiting the interior zero, the rebound and the separation of the thresholds, since the abstract stated the interior zero and the rebound at the scale level and Section 4.3 exhibited them only at a single triple. He also asked for four points of wording: the abstract's "exact vanishing is guaranteed only at $g = 1$", which reads as the uniqueness claim Proposition 3.2(iii) disclaims; the reduction after Proposition 6.1, which is to the bound (7) and not to the exact identity; the weight of Remark 6.2, defined only at states of positive probability under the mixture; and the positivity of the radicand in the proof of Theorem 4.3(i).

> **Candidate's response.** Adopted. Remark 4.5 states the attainment condition; Section 4.3 realizes it at $\lambda = 0.35$ ($g_{\mathrm{first}} = g_{\mathrm{term}} = g_1 = 0.5311$ at $\varepsilon = 0.04$, against $g_0 = 0.5555$); the abstract and the contributions paragraph state the attainment. Proposition 4.10 characterizes interior divisibility at the scale level, with the examiner's Glauber pair as instance and the constants in closed form: $M_*(\delta t) = 1/[16(1 - g^*)^2]$ for $\delta t \ge \ln 2$, a rebound peak of exactly $1/64$ at $g = (1 + g^*)/2$, and $g_{\mathrm{first}} = 0.2705$, $g_{\mathrm{term}} = 0.8721$ at $\varepsilon = 0.01$. The candidate corrected the examiner's re-entry point, 0.4885, to 0.4882. The abstract reads "exact vanishing is guaranteed at $g = 1$ and possible at no more than one interior coupling"; Remark 6.2 states the weight for states of positive probability and

completes a null column with the constant $g$, which lies between the smallest and the largest Bayes weight because both one-time laws sum to one; the other two wording points are made.

**Doctoral Examiner's disposition.** *Corrected; Proposition 4.10 re-derived in both directions, the attainment condition of Remark 4.5 correct as a conditional statement, and all figures recomputed (27 September). The examiner's figures had been numerical; the closed forms are the candidate's, including the value* $1/64$*, which the examiner had noticed and left as probably accidental. The examiner's re-entry point was computed with an interaction scale from an unresolved grid of triples (Section T.2(h)).*

### Third round (27 September): a scale-level witness for attainment

The abstract's optimality claim is a statement at the scale level, and its witness, the configuration at $\lambda = 0.35$ that the examiner supplied in the second round, lies in the pointwise example of Section 4.3, which does not realize the scale-level suprema. The examiner asked for a scale-level witness, or else a pointwise statement in the abstract, and supplied the witness: for divisible $\Gamma_a$, $\Gamma_{\mathrm{ref}}$ and $h > 0$, $\Gamma_0 = (\Gamma_a + h\Gamma_{\mathrm{ref}})/(1+h)$ is Proposition 4.10's formula at $g^* = -h$, and evaluating the identity (6) at $g = -h$ gives $X = -hK$ at every triple, so that $D_{\mathrm{CK}}(\delta t; \Gamma_g) = (1-g)(h+g)M_*(\delta t) = E(g)$ identically. He checked it on 1,140 triples with $h = 0.6$. He also asked that Remark 6.2's statement that the weight reduces to $g$ exactly when the marginals coincide be restricted to $0 < g < 1$.

**Candidate's response.** The candidate added the construction as Proposition 4.10(ii), with a paragraph reading parts (i) and (ii) as one formula with $\Gamma_a$ on either side of $\Gamma_0$ along the affine line through $\Gamma_{\mathrm{ref}}$. Remark 4.5 cites 4.10(ii) for the attainment at the scale level and labels the $\lambda = 0.35$ configuration pointwise; the abstract is unchanged. The instance with the same Glauber pair and $h = 0.6$ gives, for $\delta t \geq \ln 2$, $D_{\mathrm{CK}}(\delta t; \Gamma_g) = 0.024414(1-g)(0.6+g)$, which rises on $[0, 0.2]$ before it falls, and $g_{\mathrm{first}} = g_{\mathrm{term}} = g_1 = 0.68$, $g_0 = 0.744$ exactly at $\varepsilon = 0.01$. Remark 6.2 carries the restriction to $0 < g < 1$, and the value $1/64$ is stated for every $g^* \in (0, \frac{1}{2}]$, the range in which $\Gamma_0$ is stochastic.

**Doctoral Examiner's disposition.** *Corrected; the range* $(0, \frac{1}{2}]$ *checked for this report. The examiner verified both changes and closed the review on 28 September with nothing outstanding.*

## Further corrections identified by the candidate

**Candidate's response.** Beyond the findings, the candidate added Proposition 3.2 and Remark 3.3 (the initial rise of the exact deficit requires both $M > D_0$ and an interaction term directed against $X$); carried the quadratic envelope over to Proposition 4.8; gave the closed forms of the scale-level family, including the rebound peak $1/64$, and the range $g^* \in (0, \frac{1}{2}]$ on which that value holds; and corrected the examiner's re-entry figure.

**Doctoral Examiner's disposition.** *These corrections are in the version of record and are weighed as evidence under the governing standard.*

### Corrected architecture adopted

1. Families as abstract two-time families; the mixture as a formal fixed-weight path (Section 2, Remark 2.7).
2. The exact identity and the exact response $(1-g)S(g)$ (Theorem 3.1, Proposition 3.2).
3. Two thresholds; the quadratic envelope; attainment; scale-level families (Section 4).
4. The integrated deficit in Minkowski form (Section 5).
5. State-dependent coupling and the branch-mixture weight (Section 6).
6. Positioning and the scope of detectability (Section 7).

### Classification of results

*Sound and substantial:* Theorem 3.1 and Proposition 3.2; the two-threshold structure. *Sound, of modest depth:* Theorem 4.3, Proposition 4.8, Theorem 5.1, Proposition 6.1. *Sound, added in the course of the examination:* Proposition 4.10 and the attainment statement of Remark 4.5, from constructions the examiner proposed (23 and 27 September), with the closed forms by the candidate. *Requiring correction:* none outstanding. *Incorrect as stated (earlier versions; corrected):* the uniqueness inference of Section 7.1; the suppression claim of the abstract; the mixing remark; the process-tensor specialization.

### Correspondence (Exhibit A)

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P03-01 | 8 Sep | EX01 → CAN | Technical report transmitted |
| COR-P03-02 | 11 Sep | EX02 → CAN | Independent evaluation sent again, with her message of 4 Sep (also in COR-P01-01); forwarded by the candidate to EX01 on 11 Sep |
| COR-P03-03 | 13 Sep | CAN → EXAMINERS | Joint response to both reviews; the technical report attached for the evaluator; revised deposit |
| COR-P03-04 | 17 Sep | EX02 → CAN and EX01 | No further points; the two non-adopted recommendations accepted |
| COR-P03-05 | 23 Sep | EX01 → CAN, cc EX02 | Second round: verification record; two results proposed; four wording points |
| COR-P03-06 | 24 Sep | CAN → EX01, cc EX02 | Both results and the four wording points adopted; re-entry point corrected |
| COR-P03-07 | 27 Sep | EX01 → CAN, cc EX02 | Third round: a scale-level witness for attainment; one rider |
| COR-P03-08 | 27 Sep | CAN → EX01, cc EX02 | Proposition 4.10(ii); revision 4 |
| COR-P03-09 | 28 Sep | EX01 → CAN, cc EX02 | Both changes verified; review closed |

## T.6 Paper 4 examination: Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation

**Manuscript.** *Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation.* SSRN, DOI 10.2139/ssrn.6922758. Version of record: revision 3.1, 28 September 2026, 38 pp. (Exhibit E; exhibits/E-paper4).

### Technical examiner's verdict

The paper is technically the most accomplished of the portfolio and meets the reproducibility standard most clearly: every reported quantity of the worked example was recovered from the text alone. Its strengths are a theorem against the author's own preferred formulation (Proposition 2.3: the continuous-path weight vanishes almost surely, so the continuous-path estimator is 0/0), the reorganization of the paper onto the finite grid, a correct and sophisticated treatment of the self-normalized estimator (no finite-$N$ mean-squared-error identity; the popular variance factorization is not the asymptotic variance, 68.6 against 30.89), and a two-tier concentration structure stated with its hypotheses. Its component theorems are standard. The link to the target process, which the paper sets out to provide, rests on a hypothesis (D1) that is neither established nor estimable, as the candidate states. The defects lay in an invalid inference opening Section 7.2, the reading of Proposition 2.3, an inconsistent calibration procedure, the account of the degeneracy under refinement, and positioning. None was fatal to the paper. The recommended repair was to identify the mechanism of the degeneracy, withdraw the inference and the calibration, and state what the estimator estimates.

> **Candidate's response.** The candidate accepted the findings of both reviews, which by his account examined the same July text and converged on the same four points, recomputed every figure in the examiner's report, and rebuilt the paper around the examiner's product identity, which he records he had not seen and which changes what the paper says in four places. He declined the examiner's second remedy and one alternative of the independent evaluator, with reasons. He answered both reviews jointly on 16 September; in revision 2.3 (18 September) he restated 72 sentences and removed the citations of two later papers of the series, with no change to any result; and he made the amendments of the second and third rounds in revisions 3 (24 September) and 3.1 (28 September). He has no counterargument to the assessment that the component theorems are standard and the originality moderate.

### Findings and responses

#### Finding 2 and Questions 1–2: The selection weight is a path density

For constant drift divergence the selection weight is the surrogate's own Euler path density raised to the temperature, so the selected measure is a product of two path densities, $Q \propto \Pi \cdot p_\nu^\lambda$, and for Gaussian pairs the precisions add. The degeneracy under refinement is a quadratic-variation mismatch, not the pointwise degeneracy of Proposition 2.3. The examiner gave $1/\sigma_{\text{eff}}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2$ as exact.

*Severity: major (it changes the paper's account of its own mechanism and of its novelty).*

> **Candidate's response.** Accepted, with a correction. Proposition 3.5 gives the identity for general surrogates, with the divergence factor; Corollary 3.6 gives the Gaussian case, with the exact interior-row relation $k_{\text{eff}}^2/\sigma_{\text{eff}}^2 = k_\Pi^2/\sigma_\Pi^2 + \lambda k_\nu^2/\sigma_\nu^2$ and the examiner's relation as its leading order, $1/\sigma_{\text{eff}}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 + O(\delta t)$; the Markov property is inherited from a Markov proposal; and $\sigma_{\text{eff}} < \sigma_\Pi$ for every $\lambda > 0$ (Kolmogorov–Szegő), so the selected measure is singular to the proposal in the continuum at every fixed temperature. Section 9.4 attributes the $\chi^2$ blow-up to the mismatch, with the Girsanov ratio weight on the same grids as control.

**Doctoral Examiner's disposition.** *Correct; the examiner accepted the correction of his formula in writing (23 September, Section T.2(i)), verified the exact relation to residual* $2.4 \times 10^{-13}$ *on Example I, with the leading-order discrepancy first order in* $\delta t$*, and re-derived Corollary 3.6 on that example from the definitions (*$\sigma_{\text{eff}}^2 = 0.499177$*,* $k_{\text{eff}} = 1.5798$*,* $\chi^2(Q\|\Pi) = 1372.4$*). The* $O(\delta t)$ *appears in the abstract, the introduction and contribution (C1) (second round) and in Table 3 (third round).*

### Finding 3 and Question 3: "The conditional structure is forced"

Section 7.2 opened by inferring from the kernel-level obstruction that the conditional structure of the transfer theorem is forced. The examiner refuted it by construction: an indivisible target, a divisible surrogate and a proposal with $Q = P_{\text{true}}$ exactly, while the $D/3$ obstruction applies in full.

*Severity: major but straightforward (an invalid inference).*

> **Candidate's response.** Accepted; the sentence is deleted and its withdrawal stated. Section 7.4 explains why the inference fails; Example 7.4 is the examiner's construction ($M = 8$, $\lambda = 1/2$; least eigenvalues 0.339 and 1.270; at the triple $(3, 5, 7)$ the direct coefficient $-0.008588$ against the composed 0.305127; no triple of 56 composes). The candidate added the sentence that delimits it: the construction sets $K_\Pi = K_{\text{target}} - \lambda K_\nu$, so zero bias is attained by presupposing the target law.

**Doctoral Examiner's disposition.** *Withdrawn (sentence) and corrected; Example 7.4 recomputed (the smallest gap over the 56 triples is 0.011, at* $(6, 7, 8)$*). The delimiting sentence states the scope of the construction correctly; neither review had stated it.*

### Finding 4 and Questions 4–5: The transfer hypothesis under refinement

Hypothesis (D1) can hold under grid refinement only if the selected quadratic variation matches the target's. The examiner proposed two remedies: a design rule, or a modification preserving the proposal's quadratic variation. In the second round he found a step of the proof of Proposition 7.5 false: it claimed that the backward recursion reaches its fixed point within a bounded number of increments, whereas its contraction factor tends to one as $\delta t \to 0$ (0.9378 to 0.9995 for $M = 50$ to 6400) and the transient occupies a fixed fraction of the horizon. He proposed a Gaussian trace argument, and wrote that the parenthetical on a random limiting quadratic variation needed a hypothesis excluding an atom at $\sigma_{\text{eff}}^2 T$ or should be dropped. In the third round he found the

positivity of the rates used in the new proof and not stated, and gave the exact condition for the step $E \succeq 0$, $\kappa = k_\Pi/\sigma_\Pi^2 + \lambda k_\nu/\sigma_\nu^2 \geq 0$, with the last coordinate vector $e_M$ as witness of its necessity for that step ($e_M^\top E e_M = \delta t\, \kappa$).

*Severity: major (bears on the paper's central promise).*

> **Candidate's response.** The first remedy is adopted as Proposition 7.5: for Gaussian pairs, the total-variation distance to a target of diffusion coefficient $\sigma^2$ tends to one unless $1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 = 1/\sigma^2$; with $\sigma_\nu = \sigma$ this condition cannot hold at $\lambda \geq 1$, the temperature of the first example. Remark 7.6 restates the open problem as a design problem over triples. The second remedy is declined: subtracting a reference kinetic term converts the weight into a Girsanov ratio and the paper into a description of an established method. Question 5: a $\chi^2$-preserving temperature schedule drives the construction to the proposal. In revision 3 the candidate replaced the proof by the trace argument, with explicit bounds: in increment coordinates the selected precision is $(1/\delta t)(uI + E)$ with $E \succeq 0$, so $C_M \preceq (\delta t/u)I$, $\mathrm{Var}_Q(V_M) \leq 2T\delta t/u^2$, and $\mathbb{E}_Q[V_M]$ lies within $O(\delta t)$ of $T/u$, whence $V_M \to \sigma_\infty^2 T$ in probability; the parenthetical now requires that the law of the limiting quadratic variation have no atom at $\sigma_\infty^2 T$. In revision 3.1 the hypothesis is $\kappa \geq 0$, "as when both rates are positive", and the proof states that $u > 0$ holds for either sign of the rates and that $\kappa \geq 0$ is necessary for $E \succeq 0$, with the examiner's witness. In correspondence (29 September) the candidate showed that the conclusion of the proposition also holds for $\kappa < 0$ in the cases the examiner had swept, and gave a structural argument (by Corollary 3.6(ii), $a_{\mathrm{eff}} \in (0, 1]$ for either sign of the rates, so the selected chain is stable), so that $\kappa \geq 0$ is a hypothesis of the proof; the paper states it as a hypothesis of the proposition and claims nothing for $\kappa < 0$.

**Doctoral Examiner's disposition.** *Corrected; the trace-argument proof checked, and the hypothesis tested on three pairs with a negative rate and $\kappa > 0$ with no violation of the bounds. The non-adoption of the second remedy is reasoned and accepted. The examiner's statement that the conclusions of the proposition fail for $\kappa < 0$ is withdrawn (Section T.2(l)): the conclusion holds for $\kappa < 0$ in every case tested, including a case outside the surrogate family with both rates negative, and $\kappa \geq 0$ is needed only for the step $E \succeq 0$ of the proof.*

### Finding 1 and Question 6: The reading of Proposition 2.3

Proposition 2.3 shows that the classical functional fails as a pointwise importance weight on diffusion samples; the paper read it more widely, as an obstruction to the classical functional and to change of measure on path space.

*Severity: repairable.*

> **Candidate's response.** Accepted. Proposition 2.3 is retitled and every reading narrowed to the pointwise weight; the classical functional on regular reference paths and the Girsanov change of measure are stated to be unobstructed, with the ratio weight's bounded $\chi^2$ in the Gaussian example as contrast.

**Doctoral Examiner's disposition.** *Corrected (restated).*

**Finding 5 and Question 11: Calibration by action minimization**

Minimizing the empirical action is inconsistent even when the truth is in the family: with the observed law equal to the proposal ($k = 1$), the minimizer is $k^* = 1 + [2\delta t \sum_m \mathrm{Var}(\xi_m)]^{-1} = 2.76$, displaced by the divergence term.

*Severity: major for Section 8.4.*

> **Candidate's response.** Accepted, with the closed form. The procedure is withdrawn; calibration is Euler quasi-likelihood M-estimation, consistent when the observed law is the surrogate's Euler chain and otherwise converging to the Kullback–Leibler projection; three parameters are distinguished ($\nu_{\mathrm{QML}}$, $k = 1.000$; $\nu^\dagger$, $k = 12.39$; the withdrawn $\nu^*$, $k = 2.76$).

**Doctoral Examiner's disposition.** *Withdrawn and corrected;* $k^* = 2.7618$ *and* $k^\dagger$ *recomputed, with the kinetic sum alone minimized at* $k = 1.000$.

**Finding 6 and Question 13: The divergence term never acts**

In the Gaussian example the divergence term is constant, so nothing in the paper showed what separates the Onsager–Machlup action from the Euler exponent.

*Severity: repairable.*

> **Candidate's response.** The candidate added a second example, the double well $b = x - x^3$ at $\lambda = 3/4$, with a deterministic benchmark obtained by propagating densities through fifty one-step operators. With the divergence term $F^* = -0.547760$, $\chi^2 = 162.1$, $\sigma^2 = 44.31$; without it $-0.637163$, 169.3, 26.60. The term moves the estimand by 0.089 and reverses the sign of the selection displacement.

**Doctoral Examiner's disposition.** *Corrected; all six figures reproduced by an independent transfer-matrix quadrature, stable under mesh refinement. The finding asked only for an example in which the divergence term acts.*

**Finding 7 and Question 14: The convergence table**

The July convergence table did not support its caption, and the paper did not say that its numbers came from an archived script.

*Severity: repairable.*

> **Candidate's response.** The study was rerun at 1,000 replications with a fixed seed, bootstrap standard errors and batch ranges; at $N = 10^4$, $\sqrt{N} \cdot \mathrm{std} = 3.95 \pm 0.14$ against the asymptote 5.56. Script, seeds and output are stated to be deposited with the preprint. The candidate could not reproduce two of the examiner's Monte Carlo figures: 10,000 replications gave him $4.02 \pm 0.05$, with a value at or below 3.44 at bootstrap probability under $3 \times 10^{-4}$, and 7.53 lay above the maximum of his 200 batches of fifty (6.88). On 24 September he showed that the examiner's third run and his are compatible (1.5 standard errors of the difference), so that of the gaps between the three estimates only the gap between 3.44 and 4.02 exceeds sampling noise.

**Doctoral Examiner's disposition.** *Corrected. The examiner's third run (23 September; 1,000 replications, with a seed belonging to neither) gave* $3.817 \pm 0.127$*; he accepted 7.53 as an artefact of his implementation and left 3.44 undecided. An independent run for this report agrees with the table, and both figures are withdrawn (Section T.2(j)). The conclusions of the finding hold in all runs.*

### Finding 8 and Question 8: Positioning against path reweighting

Donati–Hartmann–Keller and Adib reweight externally generated path ensembles by a difference of two actions; the paper did not engage them, and drew distinctions from optimization and misspecification literatures that did not hold.

*Severity: expositional (attribution).*

> **Candidate's response.** Accepted. Both are cited; Section 10.1 positions the construction (a single action, hence a product rather than a ratio of densities); the distinctions of Sections 10.2–10.4 that did not hold are withdrawn, including a PAC-Bayes transfer; Table 3 is rebuilt. After the independent evaluator's comment of 16 September, the novelty claim acknowledges that products of densities are established (Hinton's product of experts) and rests on the exact product identity, its finite-grid implementation and its consequences.

**Doctoral Examiner's disposition.** *Corrected. The novelty claim of Section 10.1 now rests on the product identity of Finding 2, developed in the course of the review. The version of record credits both reviewers in one Acknowledgments sentence and carries no itemized attribution (letters of 18 and 24 September).*

### Presentational items and Questions 7, 9, 10, 12, 15, 16

The items concerned the origin of the divergence term (Question 7), the contributions (Question 9), uniform integrability in Theorem 4.1(iii) (Question 10), grid and convention dependence (Questions 12, 15), the thresholds in Remark 6.2, and the temperature schedule and the estimand (Question 16).

*Severity: expositional.*

> **Candidate's response.** The divergence term is the Itô–Stratonovich correction of the pathwise form, not a term of the Girsanov density; contributions are stated in two registers; no primitive condition for uniform integrability is supplied, and the paper says so; Section 9.6 states that the grid, the temperature, the proposal and the discretization convention are part of the estimand; the thresholds are labelled heuristics; the continuum problem is closed (the selected measures converge, to a limit singular to the proposal's, with Richardson limit $-0.84476$, while the fixed-proposal estimator has no nondegenerate limit, joint limits with $N$ growing with $M$ and adapted proposals being stated as not excluded). On Question 16 the decision-theoretic value of $F^*(\nu)$ is stated to be application-specific.

**Doctoral Examiner's disposition.** *Corrected. The answer to Question 16 is recorded as a limitation: without an externally established truth link, the decision-theoretic value of* $F^*(\nu)$ *is application-specific.*

**Independent evaluation (11 September, items 1–46; comments of 16 September)**

The independent evaluator raised, among others: the significance statement's claim of uncertainty quantification for the true system; the scope of Proposition 2.1; formula (7) as not the Euler log-density; a sharper $\chi^2$ bound for Lemma 3.9; the generalized-Bayes reading; a general refutation of the "forced" inference (take the proposal proportional to $w^{-1}P_{\mathrm{true}}$); the ill-posedness of action calibration in $\sigma_\nu$ and the KL-projection reading of quasi-likelihood; the two settings of the transfer theorem; the Mori–Zwanzig section (items 26, 31); and the PAC-Bayes and generator-based distinctions. Her comments of 16 September addressed the candidate's explanations, not the revised text, and raised three points: the novelty should attach to the exact product identity, its finite-grid implementation and its consequences, since multiplication of densities and the product-of-experts principle are established; the white-noise intensity is the two-sided or symmetrized integral of the autocovariance, the one-sided integral being not in general symmetric and positive semidefinite; and "the estimator has no limit" should be restricted to the fixed-proposal importance-sampling representation.

*Severity: major for the significance statement, the "forced" inference and the Mori–Zwanzig section; repairable otherwise.*

> **Candidate's response.** Accepted throughout on 16 September; Lemma 3.9 carries her bound; settings (K) and (L) are defined; the Mori–Zwanzig closure is written out, with the fluctuating force as white noise; her alternative presentation of the Mori–Zwanzig section was not needed once the closure was written out. On her three points (24 September, revision 3): Section 10.1 states that products of densities are established (Hinton's product of experts) and grounds the novelty in Proposition 3.5, its finite-grid implementation and its consequences; Section 8.3, which by his account already defined the intensity as the two-sided integral $\sigma\sigma^\top = \int_{\mathbb{R}} \mathbb{E}[f(t)f(0)^\top]\,dt$ (his letter of 16 September omitted the limits), now also writes it as $\int_0^\infty (R_f(t) + R_f(t)^\top)\,dt$ and states that the one-sided integral alone is neither symmetric nor positive semidefinite in general; and Sections 1 and 11 read "the fixed-proposal estimator has no nondegenerate limit".

**Doctoral Examiner's disposition.** *Corrected. The examiner found the replies to her three points well directed (28 September).*

**Second round (23 September): two amendments**

The examiner's assessment was favourable in substance. He accepted the candidate's correction of his Finding 2 (Section T.2(i)); re-derived the new material from the definitions (Corollary 3.6, the calibration, Lemma 8.1, Example 7.4 and Example II); ran the Monte Carlo study a third time (Finding 7); accepted the non-adoption of the second remedy; and asked for two amendments: a uniform argument for Proposition 7.5 (Finding 4) and the $O(\delta t)$ remainder in the abstract. He recorded that the candidate's correction of the July Mori–Zwanzig description had not been prompted by either review, and that Remark 7.6 keeps the one surviving dependency, on Paper 2, and drops those on later papers. He noted that the display of $K_\Pi$ in Example 7.4, read as extracted text rather than as the typeset page, places $\varepsilon$ inside the factor $1/\delta t$; the typeset page is correct.

> **Candidate's response.** Both made in revision 3 (24 September), with every section and result number unchanged: the proof of Proposition 7.5 (Finding 4); and the $O(\delta t)$ in the abstract, in the

introduction and in contribution (C1). The candidate reconciled the examiner's re-derived figures with the archived numerics: 24 of 26 agree to every printed digit; 20.3342 is the asymptotic variance at $k = 2.76$ and the archive's 20.3166 at the minimizer 2.7618; the two interior increment variances differ in the fourth significant digit according to the rows averaged.

**Doctoral Examiner's disposition.** *Corrected; certified against revision 3 (28 September).*

### Third round (28 September): three amendments

The examiner verified revision 3: the new proof of Proposition 7.5 replaces the recursion step entirely, the random-limit statement is correct, and the replies to the independent evaluator are well directed. He withdrew, on verification, his observation that the backward recursion yields variances conditional on the future (Section T.2(k)). He asked for three amendments: the hypothesis of Proposition 7.5 (Finding 4); the $O(\delta t)$ remainder in Table 3; and a qualification of the claim in Section 10.1 that the Girsanov weights "remain bounded in chi-square under refinement". That claim does not follow, since a likelihood ratio lies in $L^1$ of the reference measure and not necessarily in $L^2$ (on $[0, 1]$, $dQ/dP = \frac{1}{2}x^{-1/2}$ has infinite second moment); boundedness holds for the Gaussian control of Section 9.4 and requires in general a uniform second-moment condition on the densities, which for the stochastic exponential is a Novikov-type condition on $\int \theta^2 dt$. The third amendment concerned material that predated the round.

**Candidate's response.** All three made in revision 3.1 (28 September), with nothing else changed and every section and result number unchanged: the hypothesis $\kappa \geq 0$ (Finding 4); $+O(\delta t)$ in Table 3; in Section 10.1, bounded chi-square restricted to the Gaussian control of Section 9.4, with the $L^1/L^2$ distinction, the second-moment condition and the Novikov-type condition; and, in the $\chi^2$ row of Table 3, the same restriction and the second-moment condition.

**Doctoral Examiner's disposition.** *Corrected. The examiner checked Corollary 3.6 statement by statement, its claims not using the sign of the rates, and closed the paper with no outstanding amendments (28 September). On 29 September he accepted the candidate's correction on the scope of $\kappa \geq 0$ (Finding 4).*

### Further corrections identified by the candidate

**Candidate's response.** Beyond the findings, the candidate corrected the bibliographic attribution of "Stochastic optimal control matching" (to Domingo-Enrich et al.); stated that the July Table 1 and plug-in value came from an archived script and seed that the paper had not disclosed; added the Lemma 8.1 certificate for the second example, which fixes its temperature at $\lambda = 3/4$; and corrected the July description of the Mori–Zwanzig closure, which dropped the fluctuating force and so left $\sigma = 0$, not a surrogate under Definition 3.3.

**Doctoral Examiner's disposition.** *These corrections are in the version of record and are weighed as evidence under the governing standard.*

### Corrected architecture adopted

1. The classical action and its pointwise degeneracy, narrowly read (Section 2).
2. The finite-grid construction; the product identity and its Gaussian corollary (Section 3).
3. Standard SNIS theory specialized: consistency, CLT, two tiers, diagnostics (Sections 4–6).
4. Settings (K) and (L); the conditional transfer; the obstruction; Example 7.4; the design rule; the open problem (Section 7).
5. The (P2) certificate; the Mori–Zwanzig surrogate; quasi-likelihood calibration (Section 8).
6. Two worked examples with deposited code; grid dependence (Section 9).
7. Positioning (Section 10); conclusions and the closed continuum problem (Section 11).

### Classification of results

*Sound and substantial:* the finite-grid construction; Proposition 2.3; the two-tier structure and Remark 4.2; Proposition 7.2 and its independence from the selection bias; Lemma 8.1; Examples I and II. *Sound, added in the course of the examination:* Proposition 3.5, Corollary 3.6, Example 7.4, Proposition 7.5 (under $\kappa \geq 0$). *Limitation stated by the paper, not an error:* the estimand is the selected measure; its relation to a target holds only under (D1), attainable under refinement only for designed triples. *Incorrect as stated (earlier versions; withdrawn or corrected):* the "forced" inference; action-minimizing calibration; the Mori–Zwanzig closure with $\sigma = 0$; the PAC-Bayes transfer; the first proof of Proposition 7.5; the claim of bounded chi-square for Girsanov weights in general.

### Correspondence (Exhibit A)

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P04-01 | 11 Sep | EX02 → CAN | Independent evaluations of Papers 4 and 5 transmitted; forwarded by the candidate to EX01 on 11 Sep |
| COR-P04-02 | 14 Sep | EX01 → CAN | Technical report transmitted |
| COR-P04-03 | 16 Sep | CAN → EXAMINERS | Joint response to both reviews, with both reports attached; revised deposit |
| COR-P04-04 | 16 Sep | EX02 → CAN | Three remaining points on the response; forwarded by the candidate to EX01 on 17 Sep |
| COR-P04-05 | 18 Sep | CAN → EXAMINERS | Addendum on Papers 4 and 5: revision 2.3 (wording only); citations limited to preceding papers |
| COR-P04-06 | 23 Sep | EX01 → CAN and EX02 | Second round: correction of Finding 2 accepted; verification; third Monte Carlo run; two amendments |
| COR-P04-07 | 24 Sep | CAN → EX01, cc EX02 | Revision 3: the two amendments and the evaluator's three points |

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P04-08 | 28 Sep | EX01 → CAN, cc EX02 | Third round: three amendments; observation on the recursion withdrawn |
| COR-P04-09 | 28 Sep | CAN → EX01, cc EX02 | Revision 3.1 with the three amendments |
| COR-P04-10 | 28 Sep | EX01 → CAN, cc EX02 | Amendments verified; paper closed; a sweep for $\kappa < 0$ |
| COR-P04-11 | 29 Sep | CAN → EX01, cc EX02 | The conclusion of Proposition 7.5 holds for $\kappa < 0$ in the sweep |
| COR-P04-12 | 29 Sep | EX01 → CAN, cc EX02 | Correction accepted: $\kappa \geq 0$ is a hypothesis of the proof |

## T.7 Paper 5 examination: Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit

**Manuscript.** *Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit.* SSRN, DOI 10.2139/ssrn.7119779. Version of record: revision 2.1, 18 September 2026, 23 pp. (Exhibit F; 01_candidate_submission).

**Text examined in the first round.** The examiner's first-round report examined the text of 14 July 2026, not the later July deposit then current, which the independent evaluator reviewed (Section T.2(a)). Findings below are reported as the candidate answered them, against the later deposit and the version of record; the verification of the new material refers to revision 2.1.

### Technical examiner's verdict

The paper is the most conceptually ambitious of the portfolio and contains its best single idea. Lemma 3.2 shows that re-dressings act on the deficit by conjugation and never annihilate it, which suggests minimizing a flatness distance over the gauge orbit. Theorem 3.3 shows that this infimum is zero for every family: an explicit one-parameter family of re-dressings drives every family to the flat locus while preserving non-negativity and column sums, so no continuous orbit-invariant vanishing on the flat families exists (Corollary 3.4). The paper builds a metric projection theory instead, and the construction is sound: compact semialgebraic model sets, nested by a lumpability-preserving splitting, a profile that is attained and 1-Lipschitz, and a weight-independent zero stratum. The coherence identity (Proposition 2.2) is correct and useful. The defects lay in what the invariant was read to mean (a closure, so a vanishing profile is approximability, not realization), in a one-sided obstruction read as a floor, and in the numerical section, the most serious of the three, which offered upper bounds from local optimization as evidence of positive profiles, could not certify minimality of a rank, and did not meet the reproducibility standard. None was fatal to the paper; the numerical section could not stand as written. The recommended repair was to use the paper's own lower-bound machinery, or exact arithmetic, in place of the computations.

> **Candidate's response.** The candidate pointed out that the report examined a superseded text, and answered every finding against the later deposit. He withdrew the numerical study entirely and replaced it with five exact examples proved by hand; adopted the examiner's argument for stabilization at $D_{\max}$, which he records explains most of the other changes; adopted the joint-law obstruction with a change of name ($\kappa_J$) and one qualification; and corrected one item of the examiner's verification record. He has no counterargument to the independent evaluator's recommendation of major revision for calibration and positioning, or to the examiner's assessment that the individual results are of low to moderate weight and that the value lies in the architecture and the negative theorems.

### Findings and responses

#### Theorem 5.7, final clause: stabilization at D_max

A false final clause of Theorem 5.7 (on the closure of the process-induced families) had been deleted in July and termination posed as an open question. The examiner supplied an argument proving

it: every process-induced family lies in $\kappa(\Theta_D^+)$ with $D \le D_{\max} = \sum_{k=0}^{m} n^{k+1}$, hence in the closed set $F_{D_{\max}}$, so the closure of the process-induced families lies there.

> **Candidate's response.** Adopted as Theorem 5.7(d): $F_d = F_{D_{\max}}$ for every $d \ge D_{\max}$, the union of the model sets is closed and equals the closure of the process-induced families, the profile is constant from $D_{\max}$ on, and the memory rank is finite exactly on that closure. The former definition of an asymptotic liftability class becomes Corollary 5.9 (the class is empty); the threshold rule of Remark 4.7 extends to infinite rank; the abstract states the stabilization as a theorem.

**Doctoral Examiner's disposition.** *Correct; the chain of Theorem 5.7(d) and Corollary 5.9 re-derived, with $D_{\max} = 62$ for $n = 2$, $m = 4$. The examiner recorded the stabilization as a strengthening of the repair he had proposed, and the emptiness of the asymptotic liftability class as a better outcome than the open question it replaces (22 September).*

### Finding 2 and Questions 1–2: A vanishing profile is approximability

$F_d$ is a closure, so $h_d = 0$ means approximability by compressions, not realization; the examiner exhibited a witness, $M = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, with $h_2 = 0$ and no lift in $\Theta_2^+$, and proposed a compactified class whose exact image is $F_d$.

*Severity: major (the reading of the invariant).*

> **Candidate's response.** The terminological repair had been made in July ("closed-model rank"). The revision characterizes the gap (Remark 4.9: $V \in F_d$ lies outside $\kappa(\Theta_d^+)$ iff every approximating sequence of positive lifts has a block mass tending to zero), answers Question 1 at $d = n$ in both classes, prints the witness as Example 4.10, and poses the compactified class as Open Problem (vii) rather than constructing it.

**Doctoral Examiner's disposition.** *Corrected and qualified; Remark 4.9 checked, and Example 4.10 checked analytically, including* $\operatorname{dist}(V, \kappa(\theta_\varepsilon)) = 2\sqrt{3}\,\varepsilon$ *on the three-point grid. The non-adoption of the compactified class is accepted; the closed-model reading is operative throughout.*

### Finding 3 and Questions 7–8: The marginal-consistency obstruction

Theorem 5.6 tests only pairwise consistency of one-time marginals, so it is necessary and not sufficient for process-inducedness; the text first examined inferred that it "predicts" a floor for a family asserted to be process-free. The examiner proposed a joint-law obstruction strengthening it.

*Severity: major (an invalid inference, repaired in July); the strengthening is an addition.*

> **Candidate's response.** The inference was removed in July and the constant sharpened. The joint-law obstruction is Definition 5.10 and Theorem 5.11, named $\kappa_J$: a linear program that vanishes on the process-induced families and on every $F_d$, is Lipschitz, satisfies $\kappa_c \le n^2 \kappa_J$, and characterizes the process-induced families through the support of its optimal laws. The candidate qualified the examiner's "strict strengthening": the obstructions are ordered as detectors, not as floors (on Example 8.2 the floors are $1/\sqrt{2}$ and $1/4$). Example 8.3 separates them ($\kappa_c = 0$, $\kappa_J = 1/6$).

**Doctoral Examiner's disposition.** *Corrected; Examples 8.2–8.3 checked by exact linear programming ($\kappa_c = 1$, $\kappa_J = 1/4$; $\kappa_c = 0$, $\kappa_J = 1/6$), with the optimal joint laws supported on the paths printed; Theorem 5.11(c) tested on 700 random families with no violation, the constant $n^2$ being sharp and attained in Example 8.2. The qualification is right; the examiner had not drawn the distinction between detectors and floors.*

### Findings 5–7 and Questions 3–6, 9: The numerical section

Every value in Table 1 was an upper bound from multistart local optimization; establishing $h_3 > 0$ requires a lower bound, which was computed nowhere; in the text examined the table was nonmonotone (an upper bound $h_5 \leq 0.000222$ against $h_4 = 0.000000$), which Proposition 4.4 rules out for the true values and which therefore exposed optimizer error of the order of the values reported; and "code available from the author" did not meet the reproducibility standard. The remedy was the paper's own lower bound (Theorem 5.6) or exact arithmetic.

*Severity: major (material to what the section claimed, and to reproducibility).*

> **Candidate's response.** The numerical study is withdrawn — tables, optimizer, fitted points, seeds, script, certificates and data statement — because no upper bound can establish $h_{d-1} > 0$. Section 8 contains five exact examples, each proved by hand: the split-and-lump construction in general form (Example 8.1: the memory rank is at most the generating dimension of any lumped chain inducing the family); a reset family with profile at least $1/\sqrt{2}$ at every dimension (8.2); the family $(I, I, S)$ separating the obstructions (8.3); the class gap $h_2^{\mathrm{hom}} = \sqrt{3}$ against $h_2^{\mathrm{inhom}} = 0$ (8.4); and the gauge path in closed form, with the rate of collapse (8.5). No rank is claimed from computation; no certificate of minimal hidden dimension is offered for a realizable family, where both obstructions vanish.

**Doctoral Examiner's disposition.** *Withdrawn and replaced; the examples re-derived from the printed definitions (Example 8.4: the minimum of the squared distance is 3, attained only at $M = J$, so $h_2^{\mathrm{hom}} = \sqrt{3}$; Example 8.5 to $4.9 \times 10^{-11}$). Question 3 remains open; the paper implies it without stating it (second round, below). The class exchange rate of the withdrawn section survives only as Example 8.4, on a single family.*

### Finding 1 and Question 12: The gauge route

The invariant-theoretic language of Remark 3.5 (instability, quotient) was not proved; the redressing class was not identified; the examiner identified it with the similarity transformations relating quasi-realizations in hidden Markov realization theory.

*Severity: expositional.*

> **Candidate's response.** Remark 3.5 is an analogy and claims no quotient or ring of invariants; Definition 3.1 acts on the affine space; Remark 3.6 records the examiner's identification, the two extremes of the class, and a proof that the bounded-conditioning infimum is not constant on orbits, with the rate in Example 8.5. The candidate rebutted the independent evaluator's description of the class as "nonphysical", citing that identification.

**Doctoral Examiner's disposition.** *Corrected.*

**Finding 8 and Questions 13–15: Positioning and the abstract**

The paper's novelty sentence ("the metric, temporal member" of the realization family) needed positioning against approximate realization of hidden Markov models (Finesso–Grassi–Spreij), distances to model varieties, and spectral approximation of weighted automata.

*Severity: repairable (attribution).*

> **Candidate's response.** The sentence is removed. A positioning paragraph distinguishes the profile from Finesso–Grassi–Spreij (divergence rate on the full process, over HMMs of assigned order) and places it against Casanellas et al., Kubjas–Robeva–Sturmfels, Cohen–Rothblum, Balle et al., the realization and order-estimation literature, and the marginal problem (Vorob'ev, Kellerer). Remark 5.12 relates the memory rank to realization orders and claims no inequality with Hankel or nonnegative rank.

**Doctoral Examiner's disposition.** *Corrected. Further prior-art items identified by the examiner were not transmitted during the examination and are suggested comments (Report, Section 5.2).*

**Questions 10, 11, 16–18 and presentational items**

The questions concerned the Moreau decomposition at the projection (Question 10), the bound of Proposition 7.1 (Question 11), higher coherence (Question 16), rank estimation (Question 17), and refinement of the grid (Question 18).

> **Candidate's response.** Questions 10–11 concerned errors repaired in July (Clarke cone in place of Moreau; a common Frobenius lower bound in Proposition 7.1). Question 16: no higher content; the twisted coboundary satisfies $(\delta_V^2\Psi)(t,s,r,q) = \Psi(t,q)\Delta_V(q,s,r) - \Delta_V(t,r,q)\Psi(r,s)$, the analogue of $d_A^2 = F_A$. Question 17: the plug-in rank is biased upward; a threshold rule $\hat{d}_N = \min\{d : h_d(\hat{V}_N) \le \tau_N\}$ with $\tau_N/\varepsilon_N \to \infty$ is consistent (Remark 4.7). Question 18: Proposition 4.11: the rank cannot decrease under refinement.

**Doctoral Examiner's disposition.** *Corrected; Proposition 4.11 (within its qualification for homogeneous lifts with constant index gaps) and the repaired bound of Proposition 7.1 checked; the threshold rule of Remark 4.7 makes explicit what consistency of an integer-valued rank requires.*

**Independent evaluation (11 September, on the later July deposit; comments of 21 September)**

The independent evaluator, citing the July deposit throughout, asked for the scope of "without a generator", the quantum analogy stated as formal, flatness as terminology, broader positioning in six areas, the pairwise character of the marginal-consistency obstruction (and that $\kappa_c = 0$ does not give a joint law), the two-time limits of the cochain language, the Moreau passage, and certification of the numerical claims (chart stationarity, interior points, block masses), and recommended major revision for calibration and positioning. Her comments of 21 September addressed the candidate's explanations, not the revised text. She found that the changes described appear directed at her main

concerns, and that the uniform dimension bound and the closedness argument, as described, appear to supply what her compactification comment had found necessary, leaving the exact-realization and boundary questions separate. She accepted the established role of the re-dressings in quasi-realization theory, her concern having been that they need not be invertible probability-preserving changes of coordinates, so that the trivialization stays relative to the declared class; found the joint-law obstruction a direct response to her point on pairwise consistency; accepted the withdrawal of Section 8, her objection having been to the strength of the conclusions drawn from optimizer outputs, with the replacement examples requiring their own verification; withdrew her request for the supporting archive as no longer applicable; and found the expanded positioning consistent with her recommendations.

*Severity: major (numerical certification; positioning).*

> **Candidate's response.** Accepted; the numerical claims are withdrawn with the section, and the positioning paragraph meets her request. Her objection that a compactification would not by itself close the union is accepted and made moot by Theorem 5.7(d).

**Doctoral Examiner's disposition.** *Corrected. The examiner re-derived the replacement examples from the printed definitions (22 September).*

### Second round (22 September, two letters) and closure (28 September)

In his first letter of 22 September the examiner acknowledged that his report had examined a superseded text, and corrected his reading of Remark 2.3 (Section T.2(a),(b)). He recorded what the revision corrects, re-derived the new material (above), and recorded three limits of the present results: no certificate of minimal hidden dimension for an arbitrary non-flat process-induced family, so that Question 3 stands open; the converse of the joint-law obstruction when every optimal law has a vanishing marginal; and the compactified class posed and not constructed. He suggested notes of attribution at the results the review contributed to, leaving the choice to the candidate, and asked for a cross-reference to Paper 6, which he wrote settles Proposition 5.3's exponent at $d = n$ and Open Problems (ii) and (v). In his second letter the same day, having read the addendum of 18 September, he established from six markers that his verification refers to revision 2.1, and withdrew the cross-reference request as put: citing only the preceding papers is the better arrangement, a reciprocal citation making the dependence circular. He suggested a pointer in Open Problem (v), or in the presentation of the portfolio. He noted, with no request, that the candidate had withdrawn the clause that the metric charge does not decompose algebraically, which no result of the paper proves and neither report had challenged.

> **Candidate's response.** The candidate restated the three limits on 24 September, the second as the converse of Theorem 5.11(a). The paper states the second (Remark 5.12) and the third (Open Problem (vii)); it implies the first, the obstructions bounding the profile only outside the closure and Open Problem (iii) asking for sharp rank bounds, without stating it. The paper keeps one Acknowledgments sentence and no notes at the results, cites the four papers that precede it and none that follows, and leaves Proposition 5.3's exponent and Open Problems (ii) and (v) as it states them. On 28 September he undertook to record the deficit-level answer described below in the memory-production paper, since that paper cites this one, and in his next letter announced revision 2.2 of that paper, which by his account records it (COR-P06-08; Section T.8).

**Doctoral Examiner's disposition.** *No further correction required; the paper was closed with no outstanding amendments on 28 September. The examiner verified the candidate's statements against revision 2.1 and corrected his own record twice (Section T.2(m)). The residual gap is the converse of Theorem 5.11(a) when every optimal law has a vanishing marginal; part (d) closes the case in which some optimal law has positive one-time marginals. Open Problem (v) asks for a decomposition of the metric charge, which Paper 6 expressly declines at the charge level (its Corollary 4.8 gives bounds and states that the charge satisfies no additive identity of its own), so the problem is correctly posed as open. At the deficit level, which Open Problem (v) also describes, Paper 6 proves the decomposition (Theorem 4.2(i)–(ii), with the zero set of production in Theorem 4.5). The candidate restated three of the five limits in the examiner's record; the other two stand unchanged and unobjected: the empirical content, met vacuously since the paper contains no computation, and the two results not adopted, with reasons. The letter of 28 September carries forward one substantive item, the deficit-level answer to Open Problem (v); the links on Proposition 5.3's exponent and Open Problem (ii) were not taken up again.*

## Further corrections identified by the candidate

> **Candidate's response.** Beyond the findings, the candidate identified the sign error in Remark 2.3 (corrected in July, and in the examiner's record), the repairs of July (the Moreau passage, Proposition 7.1, and the false clause of Theorem 5.7), the formula for $\delta_V^2$, and the detector-versus-floor qualification of $\kappa_J$; and in revision 2.1 he withdrew the unproved clause that the metric charge does not decompose algebraically.

**Doctoral Examiner's disposition.** *These corrections are in the version of record and are weighed as evidence under the governing standard.*

## Corrected architecture adopted

1. The deficit as a 2-cochain; the coherence identity; the twisted coboundary and its square (Section 2).
2. The gauge route and its trivialization (Section 3).
3. Model sets, profile and memory rank; the closure gap and its witness; grid monotonicity (Section 4).
4. Two linear-programming obstructions; realization of process-induced families and stabilization at $D_{\max}$ (Section 5).
5. First-order conditions at the projection (Section 6); class dependence (Section 7).
6. Five exact examples (Section 8); open problems (Section 9).

## Classification of results

*Sound and substantial:* Theorem 3.3 and Corollary 3.4; Proposition 2.2; the projection theory of Section 4; Theorem 5.6; Theorem 5.7(a)–(c); Proposition 7.1; the exact examples. *Sound, added*

*in the course of the examination:* Theorem 5.7(d), Corollary 5.9, Theorem 5.11. *Open:* the degenerate converse of Theorem 5.11(a) and the compactified class, both stated as open; a certificate of minimal hidden dimension for a realizable family, implied and not stated. *Incorrect as stated (earlier versions; withdrawn or corrected):* the numerical section; the "predicts" inference; the novelty sentence; and, before the examination, the final clause of Theorem 5.7, the Moreau passage, Proposition 7.1's bound and Remark 2.3's sign.

## Correspondence (Exhibit A)

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P05-01 | 11 Sep | EX02 → CAN | Independent evaluations of Papers 4 and 5 transmitted; forwarded by the candidate to EX01 on 11 Sep (the same message as COR-P04-01) |
| COR-P05-02 | 14 Sep | EX01 → CAN | Technical report transmitted |
| COR-P05-03 | 17 Sep | CAN → EXAMINERS | Joint response to both reviews, with both reports attached; the two reviews examined different texts; one item of the examiner's record corrected; revised deposit |
| COR-P05-04 | 18 Sep | CAN → EXAMINERS | Addendum on Papers 4 and 5: revision 2.1; citations limited to preceding papers (the same message as COR-P04-05) |
| COR-P05-05 | 21 Sep | EX02 → CAN | Comments on the candidate's responses for Papers 2, 5 and 6, as forwarded to EX01 on 24 Sep (the same message as COR-P02-04) |
| COR-P05-06 | 22 Sep | EX01 → CAN, cc EX02 | Second round: the text examined and Remark 2.3 acknowledged; verification of the new material; three limits; two requests |
| COR-P05-07 | 22 Sep | EX01 → CAN, cc EX02 | Verification referred to revision 2.1; the cross-reference request withdrawn as put |
| COR-P05-08 | 24 Sep | CAN → EXAMINERS | The limits confirmed; attribution and cross-references kept as they are |
| COR-P05-09 | 28 Sep | EX01 → CAN, cc EX02 | Statements verified; two corrections of the examiner's record; paper closed |
| COR-P05-10 | 28 Sep | CAN → EX01, cc EX02 | The deficit-level answer to be recorded in Paper 6 at its next revision |

The examiner's letter of 23 September on Paper 4 (COR-P04-06) also concerns Open Problem (v) of this paper (Section T.2(m)).

## T.8 Paper 6 examination: Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit

**Manuscript.** *Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit.* SSRN, DOI 10.2139/ssrn.7206578 (from the candidate's letters). Version of record: revision 2.2, 29 September 2026, which by the candidate's account adds "one paragraph to the second paragraph of the Introduction" of revision 2.1 (24 September, 23 pp.) (Exhibit G; 01_candidate_submission) .

### Technical examiner's verdict

The paper is strong, carefully executed and correct in every checked particular, and it is the best-evidenced work in the portfolio: it contains no optimization, no simulation and no data, and every displayed quantity, verified in exact rational arithmetic, is right. It corrects its own natural conjecture before proving anything (the charge is monotone under coarse-graining in neither direction, with exact counterexamples). Lemma 2.3 is the load-bearing idea: identifying kernel differences with densities in $L^2(p_j \otimes p_i)$ makes the compression a conditional expectation onto the block $\sigma$-algebra, so that the flux is contractive and the flux/severed splitting orthogonal. Theorem 4.2 is exact, and was verified with zero failures on 200 random rational instances; Theorem 4.5 characterizes the zero set of production; Proposition 6.5 constructs a process with a strictly positive joint law whose cut severs nothing and whose production cancels the transmitted deficit, which shows that the coherence hypothesis of Theorem 6.2 is sharp. The defects lay in scope: projection-freeness was stated beyond what Theorem 5.2 proves; every charge-level statement ran through a comparison with unspecified constants and exponent, so was qualitative; the metric was placed in a class it does not belong to; "memory" was used for the two-time defect alone. None was fatal. The recommended repair was to scope the claims and, if possible, make the comparison explicit.

> **Candidate's response.** The candidate accepted the findings of both reviews, which converged on the same five points, recomputed every figure, and made the comparison explicit on every grid with exponent one (Lemma 2.5, Proposition 2.6). He declined a numerical charge column, to keep the paper in exact arithmetic. He answered both reviews jointly on 18 September (revision 2), the second round on 24 September (revision 2.1), and in revision 2.2 (29 September) recorded the deficit-level answer to Open Problem (v) of Paper 5.

### Findings and responses

#### Question 4: Explicit constants for the charge-level transfer

The examiner asked whether the constants of the comparison between the charge and the deficit norm could be bounded, for $n = 2$ on a three-point grid.

> **Candidate's response.** On every grid, with exponent one. Lemma 2.5 bounds left and right multiplication of a kernel difference by the law distortion $\varkappa(K)^{1/2}$; Proposition 2.6 gives $\|\Delta_V\|/L \le h_n(V) \le C\|\Delta_V\|$ with $L^2 = (m-1)(1+2/\rho)$ and $C^2 = (m-1)\sum_{q=0}^{m-2} \rho^{-q}$. Table 2 gives exact windows for every charged family (from Proposition 2.6 in revision 2, from Corollary 2.8 in revision 2.1); Corollary 4.8 is restated with explicit constants, its lower bound $\|\Phi + P\|/L$ no longer vacuous in the mixed regime.

**Doctoral Examiner's disposition.** *Correct; Lemma 2.5 and Proposition 2.6 re-derived. Lemma 2.5 held on* $2 \cdot 10^5$ *random triples, with worst ratios 0.938 in (a) and 0.9992 in (b), so that (b) is essentially sharp; the constants of Proposition 2.6 held on random families with* $m = 3, 4, 5$*; the windows of Table 2 recompute exactly. The question concerned* $n = 2$ *on a three-point grid; the result covers every grid and is the candidate's own.*

### Finding 1 and Question 1: The metric and the companion's class

Section 2.1 stated that the law-weighted metric (6) belongs to the admissible class of Paper 5. It does not: that class weights by the source state only.

*Severity: repairable (a false sentence whose consequences survive by norm equivalence).*

> **Candidate's response.** Accepted; the sentence was false as printed. The metric is equivalent to a member of that class, with constants $(\max p_j)^{-1/2}$ and $(\min p_j)^{-1/2}$, which is all that Lemma 4.8 and Proposition 5.3 of Paper 5 need; Proposition 2.6 proves the comparison in (6) directly; Remark 2.4 says the conditional-expectation structure fails in the companion's class.

**Doctoral Examiner's disposition.** *Corrected. In the second round Remark 2.4 gained an exact process-induced instance of that failure (law metric* $1/4 = 1/4$*; source-weighted* $7/96$ *against* $9/96$*), verified exactly, in place of a percentage range that depends on the sampling law.*

### Finding 2 and Questions 2, 9: Projection-freeness

Part (i) of Theorem 5.2 is true by definition; the uniqueness statement covers block-supported stochastic sections only, which excludes Mori's linear projections and Wiener projections; the abstract and introduction read more widely.

*Severity: repairable.*

> **Candidate's response.** Accepted. Section 5.2 is retitled "The law section among admissible sections"; the abstract, introduction and section carry the hypothesis and name what the uniqueness class does not contain.

**Doctoral Examiner's disposition.** *Corrected (restated).*

### Finding 3 and Question 8: Canonicality of the residual

Remark 5.3 called the residual at a nearest flat family canonical; the examiner exhibited a family with two nearest flat families whose residuals differ.

*Severity: repairable.*

**Candidate's response.** Canonicality is claimed for $h_n$ alone. The examiner's family is Example 5.4, with the candidate's proof that the minimum squared distance is 2 and that the two are the only minimizers. After the second round the remark cites Erdős (1946) for the null set of points with several nearest points; every citation of the example says that its family carries laws but is not process-induced; and whether a process-induced family can have several nearest flat families is Open Problem (vii). His letter of 24 September reports evidence on two symmetric slices, which the paper does not print.

**Doctoral Examiner's disposition.** *Corrected and qualified; the minimum and minimizers checked by hand. In closing (29 September) the examiner checked the Erdős attribution against two independent sources.*

### Finding 4 and Question 7: What "memory" means

In the perfect-interference construction the two-time defect of the coarse family vanishes, but the lumped process still depends on its past: the examiner computed $P(Y_2 = a \mid Y_1 = a, Y_0 = a) = 37/60$ and $P(Y_2 = a \mid Y_1 = a, Y_0 = b) = 23/36$ against $5/8$.

*Severity: expositional (scope of a word).*

**Candidate's response.** Remark 6.8 prints the computation and states that every occurrence of "memory" in the paper refers to the two-time composition defect.

**Doctoral Examiner's disposition.** *Qualified; the conditionals verified in exact arithmetic.*

### Finding 5 and Question 3: A charge column

The examiner asked for computed charges beside the deficit norms of Table 1.

**Candidate's response.** Not adopted: the paper contains no numerical optimization, and Table 2 gives the exact window of Proposition 2.6 for every charged family; positivity of a charge is certified by exact positivity of the deficit norm.

**Doctoral Examiner's disposition.** *The non-adoption is reasoned and accepted; it keeps the paper in exact arithmetic, and the paper states what the windows establish without presenting them as computed charges. Example 5.4 supplies one charge exactly, $h_2 = \sqrt{2}$, and the examiner's own minimization for Example 3.2, $h_2 = 0.0265026$, lies inside the window.*

### Finding 6: The introduction

The introduction stated that dependence on the projection "does not arise".

*Severity: expositional.*

**Candidate's response.** It now states the three parts of Theorem 5.2 and names the projections the uniqueness statement does not cover; the identities are stated to hold in every metric, the orthogonality and contraction to be properties of (6).

**Doctoral Examiner's disposition.** *Corrected.*

### Questions 5, 6, 10–16 and presentational items

The questions concerned the description of production as non-negative (Question 5), the dilation class (Question 6), the name and scope of Corollary 4.6 (Question 10), backflow (Question 11), a related orthogonal decomposition (Question 12), plug-in testing and the long kernel (Questions 13–14), higher profiles (Question 15), and the severed term (Question 16).

**Candidate's response.** Production has no sign (Definition 4.1). The time-inhomogeneous class is declared in the abstract and Section 2.1. Corollary 4.6 is "Two-time Chapman–Kolmogorov lumpability"; weak and strong lumpability imply it and neither converse holds; writing it, the candidate found the paper had no witness for the first non-implication under its own hypothesis and supplied one (Example 4.7, a deterministic four-state chain whose observed bits are pairwise independent and whose three-time law is not Markov). Backflow is an analogy. Proposition 4.9: production does not depend on the long kernel and the coherence condition is a half-space condition on it. Question 12: the same geometric content, an orthogonal split relative to a projection onto coarse structure, in a different geometry, in which the production term has no counterpart. The bounds of Corollary 4.8 are not a test, and the paper says so; the plug-in statement holds with the laws fixed and, with estimated laws, needs a margin on the block masses. The dial of Remark 6.6 is given its exact admissible intervals. $\|\Xi\|^2$ is the mean conditional variance of the deficit density given the coarse partition.

**Doctoral Examiner's disposition.** *Corrected; Example 4.7 verified in exact arithmetic, including the rows* $\pm(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$ *of* $LV(2 \leftarrow 1)Q_1$ *on the full space (squared norm 1 in the metric (6)) against zero on* $A_1$*, and the dial intervals of Remark 6.6 recomputed (*$[-5/32, 5/16]$ *and* $[-3/32, 3/16]$*). Neither report had noticed the missing witness.*

### Independent evaluation (13 September, on the August deposit; comments of 21 September)

The independent evaluator asked that the abstract not say the charge "inherits the balance law"; that the status of the metric be distinguished; that "compatible with the law" be removed; that Lemma 2.3 name its spaces; that the severed component be described as invisible only to partition-measurable observables; that Example 3.2's positivity be attributed correctly; that the two notions of closure be separated; that Corollary 4.6 keep a two-time qualifier; and that Theorem 6.2, the dial, Section 7 and the open problems be stated at their true scope. Her comments of 21 September addressed the candidate's explanations, not the revised text. She found that the changes described appear appropriately directed at her main concerns, in particular the distinction between the exact deficit-level balance and the charge-level bounds and the reading of Theorem 6.2 as monotonicity of the deficit norm; wrote that the explicit constants would strengthen the quantitative connection, subject to verification of the new argument; accepted the renaming of Corollary 4.6 in place of her qualifier; and asked that the shared zero set in the optimization discussion stay explicitly

conditional on flat fine dynamics, two-sided comparison establishing neither identical rankings nor identical minimizers across cuts.

*Severity: repairable.*

> **Candidate's response.** Accepted, with Corollary 4.6 renamed rather than qualified (there being no established two-time formulation of weak lumpability for the qualifier to refer to); Definition 4.4 defines both notions of closure; a conclusion is added. The sentence of Open Problem (v) on the zero set opens "For a flat fine family" and states that minimizing production and minimizing the charge have the same zero set and are comparable by Corollary 4.8 but need not order the cuts identically; revision 2 already read so.

**Doctoral Examiner's disposition.** *Corrected.*

**Second round (21 September): five qualifications**

The examiner compared the response with the revised text, re-derived the principal new arguments, and found the revision favourable subject to five qualifications: Example 5.4 is not process-induced, so it shows non-uniqueness among families carrying laws, and the restricted question should be posed; for a process-induced family the kernel $V(r \leftarrow s)$ transports the laws, so the argument gives a sharper lower constant ($L^2 = 2 + 1/\min p_2$ on three points), and $C = 1$ may hold on every grid; Remark 2.4 needs an exact counterexample in place of a percentage range; at $d = n$ the charge and the deficit norm carry the same information up to constants, which he called bi-Lipschitz equivalence; and the null-set statement of Remark 5.3 is a classical fact needing neither the companion's first-order theory nor variational analysis. In discussing the constants he wrote that a family may carry laws without transporting them, Example 5.4 being a case in point. He reserved a check of Remark 2.7 against Proposition 5.3 of Paper 5.

> **Candidate's response.** The candidate made all five changes in revision 2.1 (24 September): the example qualified at every citation, with Open Problem (vii); Corollary 2.8 (for law-transporting families, $\|\Delta_V\|/(2+1/\min p_2)^{1/2} \le h_n(V) \le \|\Delta_V\|$ on three points), which narrows Table 2's windows from factors of about 2.5–3.5 to 2.2–2.8; whether the upper constant is 1 on every grid posed as Open Problem (vi), his letter reporting first-order evidence at 175 flat families (smallest nonzero singular value of the linearized deficit map at least 1), which the paper does not print; the exact instance of Remark 2.4; "comparable" at $d = n$, with the content beyond the deficit norm at $d > n$, the candidate writing "comparable" rather than "bi-Lipschitz" because the statement says nothing about moduli of continuity; and Erdős cited for the null set. He showed that Example 5.4 transports its laws ($I$ and $\sigma$ preserve the uniform law) and fails to be process-induced because no joint law has its three margins. On the reserved check, Proposition 5.3 keeps its number in revision 2.1 of Paper 5, and Proposition 2.6 gives that comparison at $d = n$ with exponent one and explicit constants.

**Doctoral Examiner's disposition.** *Corrected; Corollary 2.8's proof checked and the windows* $\sqrt{9/56375}$, $\sqrt{51/475600}$, $3/(64\sqrt{5})$ *recomputed. The examiner accepted "comparable" as the correct word and closed the paper on revision 2.1 with nothing to amend (29 September).*

### Closure (29 September): Open Problem (vi) and revision 2.2

In his closing letter the examiner reported a random search over 114 process-induced families, with no ratio $h_n/\|\Delta_V\|$ above 1 (maximum 0.98305), and described it as the more informative test of Open Problem (vi).

> **Candidate's response.** The candidate repeated the search on 120 families (maximum 0.9511) and replied that random maxima understate the supremum: families placed along the worst linear direction at flat families had reached 0.999 in his earlier check. He deposited revision 2.2, which by his account adds to the Introduction a paragraph recording that Theorems 4.2 and 4.5 answer Open Problem (v) of Paper 5 at the deficit level, that Corollary 4.8 carries the decomposition to the charge as two-sided bounds, and that the charge-level decomposition remains open, and changes nothing else. He confirmed on his own families the kernel condition $\ker L \cap A \subseteq T$, to $8 \times 10^{-15}$.

**Doctoral Examiner's disposition.** *The examiner agreed that the supremum is approached near the flat families, where random draws do not land, withdrew his description of the random search (Section T.2(n)), and ran the first-order test at the flat families: over 42 flat families with $n \in \{2, 3, 4\}$ and $m \in \{3, 4, 5\}$ the smallest singular value is at least 1.000013, and the kernel condition the first-order bound requires, $\ker L \cap A \subseteq T$, holds to order $10^{-16}$. The evidence is first-order, and the paper prints only the question. The content of revision 2.2, as the candidate describes it, agrees with the examiner's verification, and as a citation of an earlier paper of the series it creates no circularity; the examiner reserved a check of the wording against the deposited text.*

## Further corrections identified by the candidate

> **Candidate's response.** Beyond the findings, the candidate supplied Example 4.7, a missing witness for a non-implication under the hypothesis of Corollary 4.6.

**Doctoral Examiner's disposition.** *These corrections are in the version of record and are weighed as evidence under the governing standard.*

## Corrected architecture adopted

1. The law metric, its equivalences, and the explicit two-sided comparison (Section 2).
2. Non-monotonicity of the charge, with exact examples (Section 3).
3. The balance law, orthogonal conservation, the zero set of production and two-time CK lumpability (Section 4).
4. The Mori–Zwanzig correspondence and its limits; the law section among admissible sections (Section 5).
5. Tower monotonicity of the deficit norm and its sharpness (Section 6).
6. Exact examples with charge windows (Section 7); conclusion and open problems (Sections 8–9).

### Classification of results

*Sound and substantial:* Lemma 2.3; Theorem 3.1; Theorem 4.2; Theorem 4.5 and Corollary 4.6; Theorem 5.2 (ii)–(iii); Theorem 6.2 and Proposition 6.5. *Sound, added in the course of the examination:* Lemma 2.5, Proposition 2.6 and Example 4.7 (revision 2); Corollary 2.8 (revision 2.1). *Open, and stated as open:* whether the upper constant is 1 on every grid (Open Problem (vi)); whether a process-induced family can have several nearest flat families (Open Problem (vii)). *Requiring correction:* none outstanding. *Incorrect as stated (earlier versions; corrected):* the metric's membership in the companion's class; non-negative production; the canonical residual; "the charge inherits the balance law".

### Correspondence (Exhibit A)

| File | Date | From → to | Content |
|---|---|---|---|
| COR-P06-01 | 13 Sep | EX02 → CAN | Independent evaluation, the sixth of the agreed set, transmitted, with the terms of any follow-up; forwarded by the candidate to EX01 on 13 Sep |
| COR-P06-02 | 15 Sep | EX01 → CAN | Technical report transmitted |
| COR-P06-03 | 18 Sep | CAN → EXAMINERS | Joint response to both reviews; revised deposit |
| COR-P06-04 | 21 Sep | EX02 → CAN | Comments on the candidate's responses for Papers 2, 5 and 6, as forwarded to EX01 on 24 Sep (the same message as COR-P02-04 and COR-P05-05) |
| COR-P06-05 | 21 Sep | EX01 → CAN, cc EX02 | Second round: verification of the new material; five qualifications |
| COR-P06-06 | 24 Sep | CAN → EXAMINERS | Revision 2.1: the five qualifications and the evaluator's point |
| COR-P06-07 | 29 Sep | EX01 → CAN, cc EX02 | Paper closed on revision 2.1; a random search on Open Problem (vi) |
| COR-P06-08 | 29 Sep | CAN → EX01, cc EX02 | Random draws understate the supremum; revision 2.2 |
| COR-P06-09 | 29 Sep | EX01 → CAN, cc EX02 | Description withdrawn; the first-order test; the described content of revision 2.2 assessed, the wording check reserved |
| COR-P06-10 | 29 Sep | CAN → EX01, cc EX02 | The kernel condition confirmed |

## T.9 Record of revisions

The following corrections were made by the candidate in response to the examinations and are incorporated in the versions of record (Exhibits B–G), against whose text each was certified. None remains outstanding. Dispositions: *corrected* (statement or proof repaired); *restated* (claim reduced to what is proved); *withdrawn* (removed, with any successor stated as an open problem); *qualified* (hypotheses or terminology made explicit).

### Paper 1 (The Physics of the Instant)

- Central claim: completeness at a point an assumption not supplied by the construction of the state, not a proved incompleteness — *restated*.
- Definition 1 of dynamical completeness ("uniquely solvable" in the deterministic sense); the hierarchy Markov $\Rightarrow$ composition $\not\Rightarrow$ Markov — *qualified*.
- Window claim replaced by path-segment lumping and Lemma 1, with regular conditional distributions, "at least linearly" and persistence conditional on a nonzero limiting defect — *corrected*.
- The falsifiable prediction of Section 6 and its financial-series confirmation; successor the one-sided test of Lemma 1, with a declared sampling floor — *withdrawn*.
- Salecker–Wigner argument: independence from gravitation; clock bound in units of time — *corrected*.
- Spatial-to-temporal inference via the position–momentum relation — *corrected*.
- Renormalization scales separated; regulator as momentum cutoff — *corrected*.
- Wightman's theorem stated for bounded fields; time-slice axiom stated, with no inference about the sharp surface — *corrected*.
- Minimum-time language scoped to an operational obstruction (nine sentences) — *qualified*.
- Landau–Lifshitz paragraph; Motahari citation — *withdrawn*.
- Section 5 confined to the collapse-timing formulation — *restated*.
- Positioning (Butterfield, Arntzenius, Albert); the problem of admissible descriptions recorded as open, with the statement that neither constraint of the paper bounds history depth or state-space dimension — *qualified*.

### Paper 2 (Koopman Spectral Theory)

- Proposition 2.11(c): padded witness for $n > 3$ — *corrected*.
- Theorem 7.4: hypothesis (C2) on the open region; Remark 7.5 — *corrected*.
- Appendix C, Step 1b: Michael's theorem removed; relative trivialization — *corrected*.
- Theorem 5.4(iv): minimality per pair $(t, s)$ — *corrected*.
- Remark 2.8: augmentation constrained by locality or noncontextuality only — *restated*.
- Section 3.2: time-homogeneous and evolution-family regimes separated — *corrected*.

- No-generator statement scoped to the Dyson (Bochner-integrable) class — *qualified.*
- Contraction assumption at every statement of the 1/3 bound — *qualified.*
- Remark 4.7's open question closed by Proposition 4.7 (sharpness, not attained) — *corrected.*
- Priority: Dyson and Blanes et al.; Canturk and Doukas; sewing and almost-flows; Table 1 — *corrected.*
- Scope-fixing labels on Proposition 3.6, Theorem 4.5(a), Corollary 4.6; lead-in to Theorem 5.4 — *qualified.*
- Remark 5.7: gauge law of the formal generator — *qualified.*
- Table 1 row for Theorem 4.8; exclusivity clause after Proposition 4.7 — *restated.*

### Paper 3 (Convex Perturbation)

- Section 7.1: uniqueness inference replaced by Proposition 3.2(iii) — *corrected.*
- The mixture as a formal fixed-weight path; "canonical" and "observer coupling" removed; Remarks 2.7 and 6.2 — *restated.*
- Interpretive conclusions of Section 7.5; successor a scope statement — *withdrawn.*
- Abstract: suppression claim — *restated.*
- Specialization to process tensors — *withdrawn.*
- Remark 4.6 (mixing estimates) and its references — *withdrawn.*
- Theorem 4.3 with the quadratic envelope; strict inequalities at thresholds — *corrected.*
- Rate scale and rate-form question — *withdrawn.*
- Theorem 5.1 in Minkowski form — *corrected.*
- Example stated as single-triple; scale-level Proposition 4.10 — *qualified.*
- Priority: Anand–Brun, De Santis–Giovannetti, Xu–Jagadish — *corrected.*

### Paper 4 (Selection Functionals)

- "The conditional structure is forced"; successor Remark 7.6 — *withdrawn.*
- Proposition 2.3 read for the pointwise weight only — *restated.*
- Product identity and Gaussian corollary, with $O(\delta t)$ where it belongs — *corrected.*
- Mechanism of the degeneracy under refinement (Section 9.4) — *corrected.*
- Calibration by action minimization; replaced by quasi-likelihood — *withdrawn.*
- Mori–Zwanzig closure with the fluctuating force — *corrected.*
- PAC-Bayes transfer; generator-based distinctions of Sections 10.2–10.3 — *withdrawn.*
- Significance statement: uncertainty quantification under the selected measure only — *restated.*
- Proposition 7.5: trace-argument proof; hypothesis $\kappa \geq 0$ — *corrected.*

- Convergence study rerun; code and seeds stated as deposited — *corrected.*
- Example II (double well), where the divergence term acts — *corrected.*
- Theorem 4.1 hypotheses as standard sufficient conditions; uniform integrability not supplied — *qualified.*
- Remark 6.2 thresholds as heuristics — *qualified.*
- Lemma 3.9: sharper $\chi^2$ bound — *corrected.*
- Section 9.6: grid, temperature, proposal and convention as part of the estimand — *qualified.*
- Section 10.1: novelty claim; $L^1/L^2$ qualification — *restated.*
- "The fixed-proposal estimator has no nondegenerate limit" — *restated.*
- Bibliographic misattribution — *corrected.*

### Paper 5 (Memory Rank)

- Numerical study of Section 8; successor five exact examples, with the certificate of minimal hidden dimension stated as open — *withdrawn.*
- Theorem 5.7(d) and Corollary 5.9: stabilization at $D_{\max}$ — *corrected.*
- Theorem 5.11: joint-law defect, ordered with $\kappa_c$ as a detector and not as a floor — *corrected.*
- Closure gap: Remark 4.9, Example 4.10; compactification posed as Open Problem (vii) — *qualified.*
- Novelty sentence; positioning paragraph — *restated.*
- "Without a generator", the quantum analogy, flatness, "dilation class" — *qualified.*
- Remark 3.5 as analogy; Remark 3.6 — *restated.*
- Definition 5.5 pairwise only; Theorem 5.6 necessity only — *qualified.*
- Twisted coboundary and $\delta_V^2$; associahedral speculation removed — *corrected.*
- Remark 4.7: consistent threshold rule — *corrected.*
- Proposition 4.11: grid monotonicity — *corrected.*
- Remark 5.12: relation to realization orders — *qualified.*
- Significance statement; citation of Paper 6 — *withdrawn.*

### Paper 6 (Memory Production)

- Metric (6) outside the companion's class; equivalence constants — *corrected.*
- Lemma 2.5, Proposition 2.6, Corollary 2.8: explicit comparison; Corollary 4.8 restated — *corrected.*
- Abstract: the charge inherits two-sided control, not the balance law — *restated.*
- Projection-freeness and the uniqueness class of Section 5.2 — *restated.*

- "Non-negative" production — *withdrawn*.
- Remark 5.3 canonicality for $h_n$ only; Example 5.4 marked not process-induced — *qualified*.
- Corollary 4.6 renamed; Example 4.7 — *qualified*.
- Remark 6.8: "memory" as the two-time defect — *qualified*.
- Dilation class declared — *qualified*.
- Remark 2.4: exact process-induced instance — *corrected*.
- Erdős (1946) for the null set — *corrected*.
- Open Problems (vi), (vii) added; (v) conditional — *qualified*.